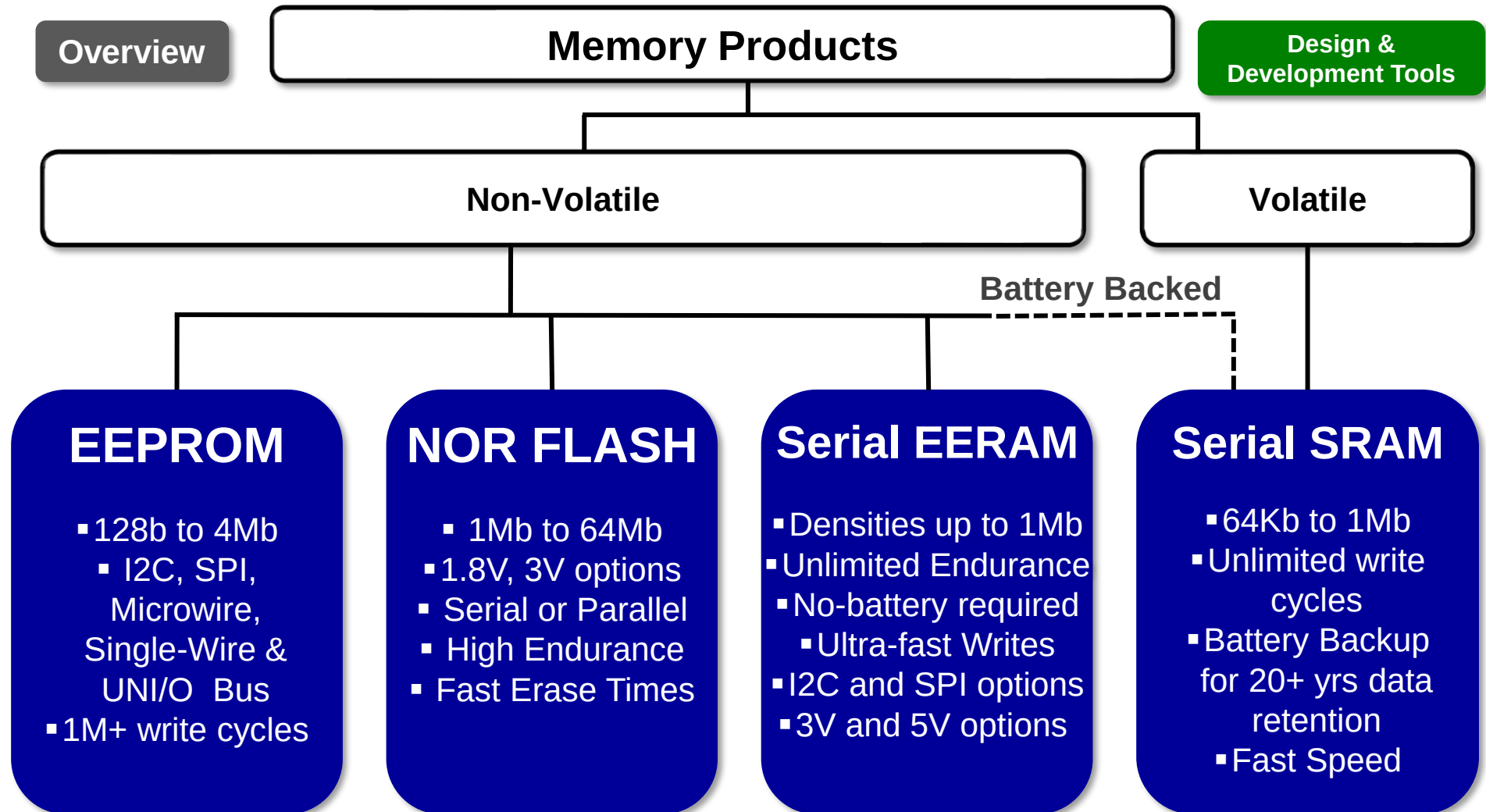
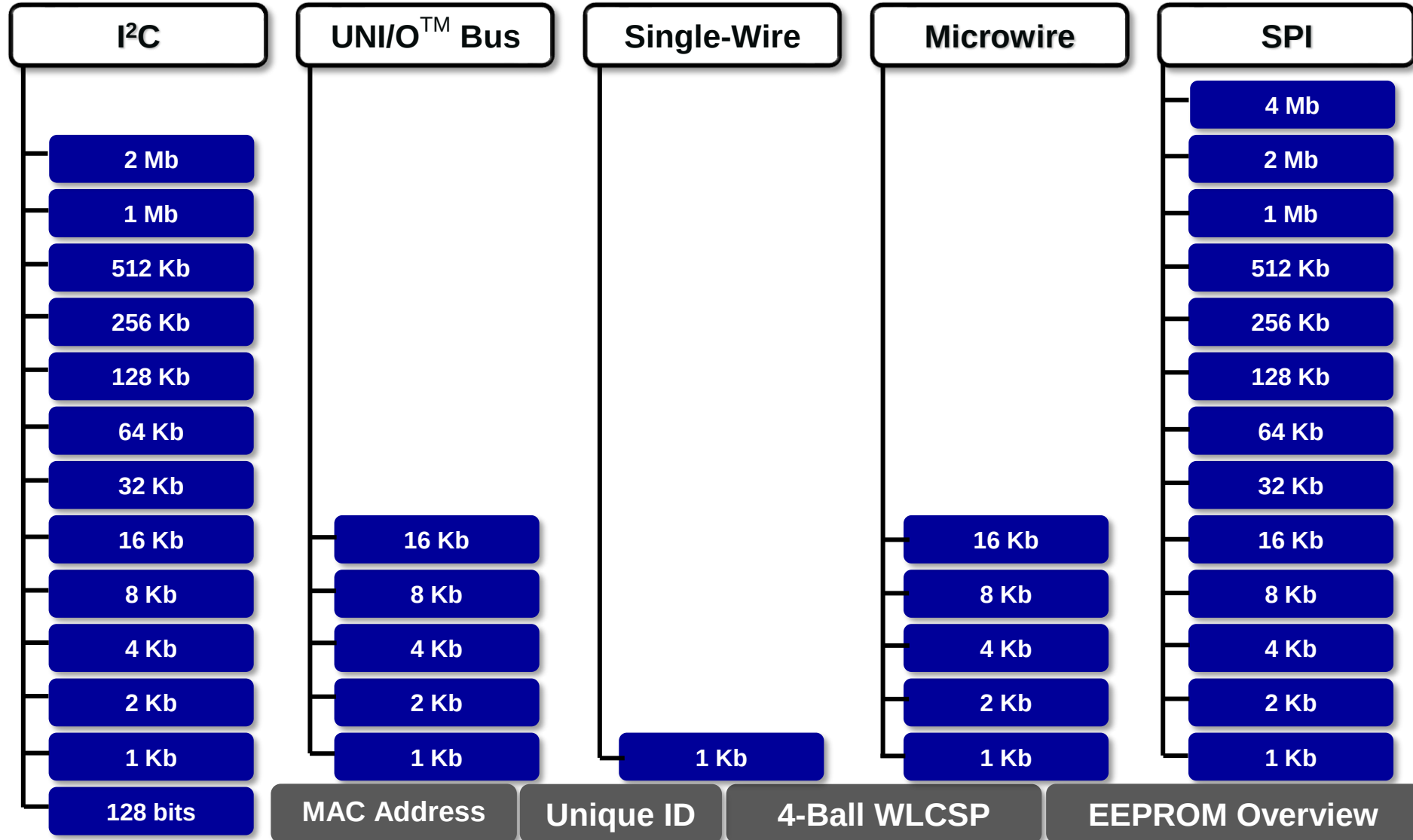


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Serial EEPROM

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128 Bit Serial EEPROMs

	24xx00
Organization	16 x 8
Operating Voltage	1.8 V to 6.0 V (24AA00) 2.5 V to 5.5 V (24LC00) 4.5 V to 5.5 V (24C00)
Max Clock Freq	400 KHz
Page Size	1 Bytes
Write Protection	None
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp, 24C00 only)
Endurance	1,000,000
Packages	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, TDFN2x3-8

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1Kb I²C Serial EEPROMs

	24xx01 	24xx014 	AT24C01C 	AT24C01D
Organization	128 x 8	128 x 8	128 x 8	128 x 8
Operating Voltage	1.7 V to 5.5 V (24FC01) 1.7 V to 5.5 V (24AA01) 2.5 V to 5.5 V (24LC01)	1.5 V to 3.6 V (24VL014) 1.7 V to 5.5 V (24AA014) 2.5 V to 5.5 V (24LC014)	1.7 V to 5.5 V	1.7 V to 3.6 V
Max Clock Freq	400 KHz 1 MHz (24FC01)	400 KHz 1 MHz (24LC014H)	1 MHz	1 MHz
Page Size	8 Bytes	16 Bytes	8 Bytes	8 Bytes
Write Protection	Half, Full Array (HW)	Half, Full Array (HW)	Full Array (HW)	Full Array (HW)
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)	-40C to 85C
Endurance	1,000,000	1,000,000	1,000,000	1,000,000
Packages	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, SC-70-5, TDFN2x3-8, UDFN2x3-8	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, UDFN2x3-8, VFBGA-8	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, UDFN2x3-8, VFBGA-8

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2Kb I²C Serial EEPROMs

	24xx02 	24xx024/5 	AT24C02C 	AT24C02D
Organization	256 x 8	256 x 8	256 x 8	256 x 8
Operating Voltage	1.7 V to 5.5 V (24FC02) 1.7 V to 5.5 V (24AA02) 2.5 V to 5.5 V (24LC02B)	1.5 V to 3.6 V (24VL024/5) 1.7 V to 5.5 V (24AA024/5) 2.5 V to 5.5 V (24LC024/5)	1.7 V to 5.5 V	1.7 V to 3.6 V
Max Clock Freq	400 KHz 1 MHz (24FC02)	400 KHz 1 MHz (24LC024H)	1 MHz	1 MHz
Page Size	8 Bytes	16 Bytes	8 Bytes	8 Bytes
Write Protection	Half, Full Array	None (24xx025), Half (24xx024H), Full Array (24xx024)	Half (AT24HC02C), Full Array (HW)	Full Array (HW)
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)	-40C to 85C
Endurance	1,000,000	1,000,000	1,000,000	1,000,000
Packages	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, SC-70-5, TDFN2x3-8, UDFN2x3-8	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, UDFN2x3-8, VFBGA-8	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, UDFN2x3-8, VFBGA-8

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4Kb I²C Serial EEPROMs

	24xx04 	AT24HC04B 	AT24C04C 	AT24C04D
Organization	512 x 8	512 x 8	512 x 8	512 x 8
Operating Voltage	1.7 V to 5.5 V (24FC04) 1.7 V to 5.5 V (24AA04) 2.5 V to 5.5 V (24LC04B)	1.8 V to 5.5 V	1.7 V to 5.5 V	1.7 V to 3.6 V
Max Clock Freq	400 KHz 1 MHz (24FC04)	1 MHz	1 MHz	1 MHz
Page Size	16 Bytes	16 Bytes	16 Bytes	16 Bytes
Write Protection	Half, Full Array (HW)	Half Array (HW)	Full Array (HW)	Full Array (HW)
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C	-40C to 85C -40C to 125C (Automotive version)	-40C to 85C
Endurance	1,000,000	1,000,000	1,000,000	1,000,000
Packages	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8, UDFN2x3-8, CSP	PDIP-8, SOIC-8, TSSOP-8	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, UDFN2x3-8, VFBGA-8	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, UDFN2x3-8, VFBGA-8

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8Kb I²C Serial EEPROMs

	24xx08 	AT24C08C 	AT24C08D
Density	1024 x 8	1024 x 8	1024 x 8
Operating Voltage	1.7 V to 5.5 V (24FC08, 24AA08) 2.5 V to 5.5 V (24LC08B)	1.7 V to 5.5 V	1.7 V to 3.6 V
Max Clock Freq	400 KHz 1 MHz (24FC08)	1 MHz	1 MHz
Page Size	16 Bytes	16 Bytes	16 Bytes
Write Protection	Half, Full Array	Full Array (HW)	Full Array (HW)
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)	-40C to 85C
Endurance	1,000,000	1,000,000	1,000,000
Packages	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8, UDFN2x3-8, CSP	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, UDFN2x3-8, VFBGA-8	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, UDFN2x3-8, VFBGA-8, CSP

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16Kb I²C EEPROMs

	24xx16 	AT24C16C 	AT24C16D
Organization	2048 x 8	2048 x 8	2048 x 8
Operating Voltage	1.7 V to 5.5 V (24FC16, 24AA16) 2.5 V to 5.5 V (24LC16B)	1.7 V to 5.5 V	1.7 V to 3.6V
Max Clock Freq	400 KHz 1 MHz (24FC16)	1 MHz	1 MHz
Page Size	16 Bytes	16 Bytes	16 Bytes
Write Protection	Half, Full Array (HW)	Full Array (HW)	Full Array (HW)
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)	-40C to 85C
Endurance	1,000,000	1,000,000	1,000,000
Packages	SOT-23-5, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8, UDFN2x3-8, CSP	UDFN-8, XDFN-8, VFBGA-8, PDIP-8, SOIC-8, SOT23-5, TSSOP-8	SOIC-8, TSSOP-8, UDFN-8, PDIP- 8, SOT23-5, VFBGA-8, CSP

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32Kb I²C EEPROMs

	24xx32A 	AT24C32D 	AT24C32E
Organization	4096 x 8	4096 x 8	4096 x 8
Operating Voltage	1.7 V to 5.5 V (24AA32A) 2.5 V to 5.5 V (24LC32A)	1.7V to 5.5V	1.7 V to 3.6V
Max Clock Freq	400 KHz	1 MHz	1 MHz
Page Size	32 Bytes	32 Bytes	32 Bytes
Write Protection	Quarter, Whole Array	Full Array (HW)	Full Array (HW)
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)	-40C to 85C
Endurance	1,000,000	1,000,000	1,000,000
Packages	SOT-23-5, SOIC-8, SOIJ-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8, CSP	UDFN-8, XDFN-8, VFBGA-8, PDIP-8, SOIC-8, SOT23-5, TSSOP-8, CSP	SOIC-8, TSSOP-8, UDFN-8, PDIP-8, SOT23-5, VFBGA-8, CSP

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64Kb I²C Serial EEPROMs

	24xx64 	AT24C64B	AT24C64D 
Organization	8192 x 8	8192 x 8	8192 x 8
Operating Voltage	1.7 V to 5.5 V (24AA64 and 24FC64) 2.5 V to 5.5 V (24LC64)	1.8 V to 5.5 V 2.7 V to 5.5 V	1.7 V to 5.5 V
Max Clock Freq	400 KHz 1 MHz (24FC64)	400KHz	1 MHz
Page Size	32 Bytes	32 Bytes	32 Bytes
Write Protection	Quarter (24xx64F) or Whole Array	Whole Array	Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)	-40C to 85C
Endurance	1,000,000	1,000,000	1,000,000
Packages	SOT-23-5, SOIC-8, SOIJ-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	TSSOP-8, SOIC-8	SOIC-8, TSSOP-8, UDFN-8, PDIP-8, SOT23-5, VFBGA-8, CSP

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128Kb I²C Serial EEPROMs

	24xx128 	AT24C128C 
Organization	16,384 x 8	16,384 x 8
Operating Voltage	1.7 V to 5.5 V (24AA128 and 24FC128) 2.5 V to 5.5 V (24LC128)	1.7 V to 5.5 V
Max Clock Freq	400 KHz 1 MHz (24FC128)	1 MHz
Page Size	64 Bytes	64 Bytes
Write Protection	Whole Array	Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)
Endurance	1,000,000	1,000,000
Packages	SOIC-8, SOIJ-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8, CSP	UDFN-8, SOIC-8, TSSOP-8, VFBGA-8, XDFN-8

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256Kb I²C Serial EEPROMs

	24xx256 	AT24C256C 
Organization	32,768 x 8	32,768 x 8
Operating Voltage	1.7 V to 5.5 V (24AA256 and 24FC256) 2.5 V to 5.5 V (24LC256)	1.7 V to 5.5 V
Max Clock Freq	400 KHz 1 MHz (24FC256)	1 MHz
Page Size	64 Bytes	64 Bytes
Write Protection	Whole Array	Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)
Endurance	1,000,000	1,000,000
Packages	SOIC-8, SOIJ-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8, CSP	VFBGA-8, UDFN-8, SOIC-8, TSSOP-8

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512Kb I²C Serial EEPROMs

	24xx512 	AT24C512C
Organization	65,536 x 8	65,536 x 8
Operating Voltage	1.7 V to 5.5 V (24AA512) 2.5 V to 5.5 V (24LC512 and 24FC512)	1.7 V to 3.6 V 2.5 V to 5.5 V
Max Clock Freq	400 KHz (24AA512 and 24LC512) 1 MHz (24FC512)	1 MHz
Page Size	128 Bytes	128 Bytes
Write Protection	Whole Array	Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C
Endurance	1,000,000	1,000,000
Packages	SOIJ-8, PDIP-8, DFN6x5-8, SOIC-8, TSSOP-8, TSSOP-14, WLCSP	SOIC-8, SOIJ-8, TSSOP-8, UDFN-8, VFBGA- 8, CSP

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1Mb I²C Serial EEPROMs

	24xx1025 	24xx1026	AT24CM01
Organization	128K x 8	128K x 8	128K x 8
Operating Voltage	1.7 V to 5.5 V (24AA1025) 2.5 V to 5.5 V (24LC1025) 1.6 V to 5.5 V (24FC1025)	1.7 V to 5.5 V (24AA1026) 2.5 V to 5.5 V (24LC1026) 1.6 V to 5.5 V (24FC1026)	1.7 V to 5.5 V
Max Clock Freq	400 KHz 1 MHz (24FC1025)	400 KHz 1 MHz (24FC1026)	1 MHz
Page Size	128 Bytes	128 Bytes	256 Bytes
Write Protection	Whole Array	Whole Array	Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C
Endurance	1,000,000	1,000,000	1,000,000
Packages	SOIJ-8, PDIP-8, SOIC-8	SOIJ-8, PDIP-8, SOIC-8, TSSOP-8	SOIC-8, SOIJ-8, TSSOP-8, CSP

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2Mb I²C Serial EEPROM

	AT24CM02
Organization	256K x 8
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	1MHz
Page Size	256 Byte
Write Protection	Full Array
Temp Range	-40C to 85C
Endurance	1,000,000
Packages	SOIC-8, CSP

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1Kb UNI/O[®] Serial EEPROM

	11xx010 
Density	1Kb (x8)
Operating Voltage	1.8 V to 5.5 V (11AA010) 2.5 V to 5.5 V (11LC010)
Max Clock Freq	100 KHz
Page Size	16 Bytes
Write Protection	Quarter, Half, Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)
Endurance	1,000,000
Packages	SOT-23-3, TO-92-3, SOIC-8, PDIP-8, MSOP-8, TDFN2x3-8

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2Kb UNI/O[®] Serial EEPROM

	11xx020 
Density	2Kb (x8)
Operating Voltage	1.8 V to 5.5 V (11AA020) 2.5 V to 5.5 V (11LC020)
Max Clock Freq	100 KHz
Page Size	16 Bytes
Write Protection	Quarter, Half, Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)
Endurance	1,000,000
Packages	SOT-23-3, TO-92-3, SOIC-8, PDIP-8, MSOP-8, TDFN2x3-8

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4Kb UNI/O[®] Serial EEPROM

	11xx040 
Density	4Kb (x8)
Operating Voltage	1.8 V to 5.5 V (11AA040) 2.5 V to 5.5 V (11LC040)
Max Clock Freq	100 KHz
Page Size	16 Bytes
Write Protection	Quarter, Half, Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)
Endurance	1,000,000
Packages	SOT-23-3, TO-92-3, SOIC-8, PDIP-8, MSOP-8, TDFN2x3-8

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8Kb UNI/O[®] Serial EEPROM

	11xx080 
Density	8Kb (x8)
Operating Voltage	1.8 V to 5.5 V (11AA080) 2.5 V to 5.5 V (11LC080)
Max Clock Freq	100 KHz
Page Size	16 Bytes
Write Protection	Quarter, Half, Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)
Endurance	1,000,000
Packages	SOT-23-3, TO-92-3, SOIC-8, PDIP-8, MSOP-8, TDFN2x3-8

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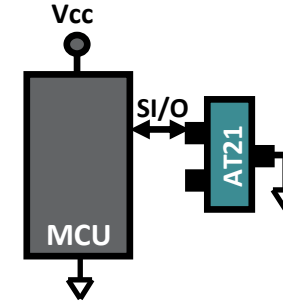
16Kb UNI/O[®] Serial EEPROM

	11xx160 
Density	16Kb (x8)
Operating Voltage	1.8 V to 5.5 V (11AA160) 2.5 V to 5.5 V (11LC160)
Max Clock Freq	100 KHz
Page Size	16 Bytes
Write Protection	Quarter, Half, Whole Array
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)
Endurance	1,000,000
Packages	SOT-23-3, TO-92-3, SOIC-8, PDIP-8, MSOP-8, TDFN2x3-8

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Single-Wire EEPROM

- True 2-pin device – SI/O and GND
- Powered through the SI/O signal
- Ideal for Identification / Counterfeit Protection / Disposables
- Pre-programmed 64-bit serial number



	AT21CSx1
Density	128 x 8 (1Kb)
Operating Voltage	1.7 V to 3.6 V (AT21CS01) 2.7 V to 4.5 V (AT21CS11)
Speed	125Kbps
Page Size	8 Bytes
Write Protection	Software 4-zones, security register
Temp Range	-40C to 85C
Endurance	1,000,000
Packages	SOT-23-3, SOIC-8, WLCSP

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1Kb Microwire EEPROMs

	93xx46A 	93xx46B 	93xx46C	AT93C46D 	AT93C46E
Density	1Kb (x8)	1Kb (x16)	1Kb (x8 or x16)	128 x 8 or 64 x 16	64 x 16
Operating Voltage	1.8 V to 5.5 V (93AA46A) 2.5 V to 5.5 V (93LC46A) 4.5 V to 5.5 V (93C46A)	1.8 V to 5.5 V (93AA46B) 2.5 V to 5.5 V (93LC46B) 4.5 V to 5.5 V (93C46B)	1.8 V to 5.5 V (93AA46C) 2.5 V to 5.5 V (93LC46C) 4.5 V to 5.5 V (93C46C)	1.8 V to 5.5 V	1.8 V to 5.5 V
Max Clock Freq	2 MHz	2 MHz	3 MHz	2 MHz	2 MHz
Write Protection	None	None	None	None	None
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)	-40C to 85C
Endurance	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000
Packages	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	PDIP-8, TSSOP-8, SOIC-8, VFBGA-8, UDFN-8	PDIP-8, SOIC-8, TSSOP-8
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2Kb Microwire EEPROMs

	93xx56A 	93xx56B 	93xx56C	AT93C56B 
Density	2Kb (x8)	2Kb (x16)	2Kb (x8 or x16)	256 x 8 or 128 x 16
Operating Voltage	1.8 V to 5.5 V (93AA56A) 2.5 V to 5.5 V (93LC56A) 4.5 V to 5.5 V (93C56A)	1.8 V to 5.5 V (93AA56B) 2.5 V to 5.5 V (93LC56B) 4.5 V to 5.5 V (93C56B)	1.8 V to 5.5 V (93AA56C) 2.5 V to 5.5 V (93LC56C) 4.5 V to 5.5 V (93C56C)	1.7 V to 5.5 V
Max Clock Freq	2 MHz	2 MHz	3 MHz	2 MHz
Write Protection	None	None	None	None
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)
Endurance	1,000,000	1,000,000	1,000,000	1,000,000
Packages	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3- 8, TDFN2x3-8	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOIC-8, TSSOP-8, UDFN-8, XDFN-8, VFBGA-8

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4Kb Microwire EEPROMs

	93xx66A	93xx66B	93xx66C	AT93C66B 
Density	4Kb (x8)	4Kb (x16)	4Kb (x8 or x16)	512 x 8 or 256 x 16
Operating Voltage	1.8 V to 5.5 V (93AA66A) 2.5 V to 5.5 V (93LC66A) 4.5 V to 5.5 V (93C66A)	1.8 V to 5.5 V (93AA66B) 2.5 V to 5.5 V (93LC66B) 4.5 V to 5.5 V (93C66B)	1.8 V to 5.5 V (93AA66C) 2.5 V to 5.5 V (93LC66C) 4.5 V to 5.5 V (93C66C)	1.7 V to 5.5 V
Max Clock Freq	2 MHz	2 MHz	3 MHz	2 MHz
Write Protection	None	None	None	None
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)
Endurance	1,000,000	1,000,000	1,000,000	1,000,000
Packages	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOIC-8, TSSOP-8, UDFN-8, XDFN-8, VFBGA-8

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8Kb Microwire EEPROMs

	93xx76A 	93xx76B 	93xx76C 
Density	8Kb (x8)	8Kb (x16)	8Kb (x8 or x16)
Operating Voltage	1.8 V to 5.5 V (93AA76A) 2.5 V to 5.5 V (93LC76A) 4.5 V to 5.5 V (93C76A)	1.8 V to 5.5 V (93AA76B) 2.5 V to 5.5 V (93LC76B) 4.5 V to 5.5 V (93C76B)	1.8 V to 5.5 V (93AA76C) 2.5 V to 5.5 V (93LC76C) 4.5 V to 5.5 V (93C76C)
Max Clock Freq	2 MHz	2 MHz	3 MHz
Write Protection	None	None	Yes
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)
Endurance	1,000,000	1,000,000	1,000,000
Packages	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8

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16Kb Microwire EEPROMs

	93xx86A 	93xx86B 	93xx86C 	AT93C86A 
Density	2048 x 8	1024 x 16	2048 x 8 or 1024 x 16	2048 x 8 or 1024 x 16
Operating Voltage	1.8 V to 5.5 V (93AA86A) 2.5 V to 5.5 V (93LC86A) 4.5 V to 5.5 V (93C86A)	1.8 V to 5.5 V (93AA86B) 2.5 V to 5.5 V (93LC86B) 4.5 V to 5.5 V (93C86B)	1.8 V to 5.5 V (93AA86C) 2.5 V to 5.5 V (93LC86C) 4.5 V to 5.5 V (93C86C)	1.8 V to 5.5 V
Max Clock Freq	2 MHz	2 MHz	3 MHz	2 MHz
Write Protection	None	None	Yes	Yes
Temp Range	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C (I-Temp) -40C to 125C (E-Temp)	-40C to 85C -40C to 125C (Automotive version)
Endurance	1,000,000	1,000,000	1,000,000	1,000,000
Packages	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8	SOT-23-6, SOIC-8, TSSOP-8, PDIP-8, MSOP-8	SOIC-8, TSSOP-8, PDIP-8, MSOP-8, DFN2x3-8, TDFN2x3-8	SOIC-8, TSSOP-8, PDIP-8, UDFN-8

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1Kb SPI Serial EEPROMs

	25xx010A 	AT25010B 
Organization	128 x 8	128 x 8
Operating Voltage	1.8 V to 5.5 V (25AA010A) 2.5 V to 5.5 V (25LC010A)	1.8 V to 5.5 V
Max Clock Freq	10 MHz	20 MHz
Page Size	16 Bytes	8 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C -40°C to 125°C (Automotive version)
Endurance	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, TSSOP-8, MSOP-8, DFN2x3-8, TDFN2x3-8, SOT-23-6	SOIC-8, TSSOP-8, UDNF-8, XDFN-8, VFBGA-8

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2Kb SPI Serial EEPROMs

	25xx020A 	AT25020B 
Organization	256 x 8	256 x 8
Operating Voltage	1.8 V to 5.5 V (25AA020A) 2.5 V to 5.5 V (25LC020A)	1.8 V to 5.5 V
Max Clock Freq	10 MHz	20 MHz
Page Size	16 Bytes	8 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C -40°C to 125°C (Automotive version)
Endurance	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, TSSOP-8, MSOP-8, DFN2x3-8, TDFN2x3-8, SOT-23-6	UDFN-8, SOIC-8, TSSOP-8, VFBGA-8

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4Kb SPI Serial EEPROMs

	25xx040A 	AT25040B 
Organization	512 x 8	512 x 8
Operating Voltage	1.8 V to 5.5 V (25AA040A) 2.5 V to 5.5 V (25LC040A)	1.8 V to 5.5 V
Max Clock Freq	10 MHz	20 MHz
Page Size	16 Bytes	8 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C -40°C to 125°C (Automotive version)
Endurance	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, TSSOP-8, MSOP-8, DFN2x3-8, TDFN2x3-8, SOT-23-6	UDFN-8, SOIC-8, TSSOP-8, VFBGA-8, XDFN-8

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8Kb SPI Serial EEPROMs

	25xx080C 	25xx080D 	AT25080B 
Organization	1024 x 8	1024 x 8	1024 x 8
Operating Voltage	1.8 V to 5.5 V (25AA080C) 2.5 V to 5.5 V (25LC080C)	1.8 V to 5.5 V (25AA080D) 2.5 V to 5.5 V (25LC080D)	1.8 V to 5.5 V
Max Clock Freq	10 MHz	10 MHz	20 MHz
Page Size	16 Bytes	32 Bytes	32 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C -40°C to 125°C (Automotive version)
Endurance	1,000,000	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, TSSOP-8, MSOP-8, TDFN2x3-8	PDIP-8, SOIC-8, TSSOP-8, MSOP-8, TDFN2x3-8	UDFN-8, XDFN-8, VFBGA-8, SOIC-8, TSSOP-8, CSP

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16Kb SPI Serial EEPROMs

	25xx160C 	25xx160D 	AT25160B 
Organization	2048 x 8	2048 x 8	2048 x 8
Operating Voltage	1.8 V to 5.5 V (25AA160C) 2.5 V to 5.5 V (25LC160C)	1.8 V to 5.5 V (25AA160D) 2.5 V to 5.5 V (25LC160D)	1.8 V to 5.5 V
Max Clock Freq	10 MHz	10 MHz	20 MHz
Page Size	16 Bytes	32 Bytes	32 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C -40°C to 125°C (Automotive version)
Endurance	1,000,000	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, TSSOP-8, MSOP-8, TDFN2x3-8	PDIP-8, SOIC-8, TSSOP-8, MSOP-8, TDFN2x3-8	UDFN-8, XDFN-8, VFBGA-8, SOIC-8, TSSOP-8, CSP

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32Kb SPI Serial EEPROMs

	25xx320A 	AT25320B 
Density	4096 x 8	4096 x 8
Operating Voltage	1.8 V to 5.5 V (25AA320A) 2.5 V to 5.5 V (25LC320A)	1.8 V to 5.5 V
Max Clock Freq	10 MHz	20 MHz
Page Size	32 Bytes	32 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C -40°C to 125°C (Automotive version)
Endurance	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, TSSOP-8, MSOP-8, TDFN2x3-8	UDFN-8, XDFN-8, VFBGA-8, SOIC-8, TSSOP-8

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64Kb SPI Serial EEPROMs

	25xx640A 	AT25640B 
Organization	8,192 x 8	8,192 x 8
Operating Voltage	1.8 V to 5.5 V (25AA640A) 2.5 V to 5.5 V (25LC640A)	1.8 V to 5.5 V
Max Clock Freq	10 MHz	20 MHz
Page Size	32 Bytes	32 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C -40°C to 125°C (Automotive version)
Endurance	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, TSSOP-8, MSOP-8, TDFN2x3-8, DFN5x6-8	UDFN-8, XDFN-8, VFBGA-8, SOIC-8, TSSOP-8

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128Kb SPI Serial EEPROMs

	25xx128 	AT25128B 
Organization	16,384 x 8	16,384 x 8
Operating Voltage	1.8 V to 5.5 V (25AA128) 2.5 V to 5.5 V (25LC128)	1.8 V to 5.5 V
Max Clock Freq	10 MHz	20 MHz
Page Size	64 Bytes	64 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C -40°C to 125°C (Automotive version)
Endurance	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, SOIJ-8, TSSOP-8, DFN6x5-8	UDFN-8, VFBGA-8, SOIC-8, TSSOP-8

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256Kb SPI Serial EEPROMs

	25xx256 	AT25256B 
Organization	32,768 x8	32,768 x8
Operating Voltage	1.8 V to 5.5 V (25AA256) 2.5 V to 5.5 V (25LC256)	1.8 V to 5.5 V
Max Clock Freq	10 MHz	20 MHz
Page Size	64 Bytes	64 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C -40°C to 125°C (Automotive version)
Endurance	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, SOIJ-8, TSSOP-8, DFN6x5-8	UDFN-8, VFBGA-8, SOIC-8, TSSOP-8

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512Kb SPI Serial EEPROMs

	25xx512 	AT25512
Organization	65,536 x 8	65,536 x 8
Operating Voltage	1.8 V to 5.5 V (25AA512) 2.5 V to 5.5 V (25LC512)	1.8 V to 5.5 V
Max Clock Freq	20 MHz	20 MHz
Page Size	128 Bytes	128 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C
Endurance	1,000,000	1,000,000
Packages	PDIP-8, SOIC-8, SOIJ-8, DFN6x5-8	UDFN-8, SOIC-8, TSSOP-8

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1Mb SPI Serial EEPROMs

	25xx1024 	AT25M01
Organization	131,072 x 8	131,072 x 8
Operating Voltage	1.8 V to 5.5 V (25AA1024) 2.5 V to 5.5 V (25LC1024)	1.7 V to 5.5V
Max Clock Freq	20 MHz	20 MHz
Page Size	256 Bytes	256 Bytes
Write Protection	Quarter, Half, Whole Array	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C (I-Temp) -40°C to 125°C (E-Temp)	-40°C to 85°C
Endurance	1,000,000	1,000,000
Packages	PDIP-8, SOIJ-8, DFN6x5-8	SOIC-8, SOIJ-8, CSP

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2Mb SPI Serial EEPROM

	AT25M02
Organization	262,144 x 8
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	5 MHz
Page Size	256 Bytes
Write Protection	Quarter, Half, Whole Array
Temp Range	-40°C to 85°C
Endurance	1,000,000
Packages	SOIC-8, CSP

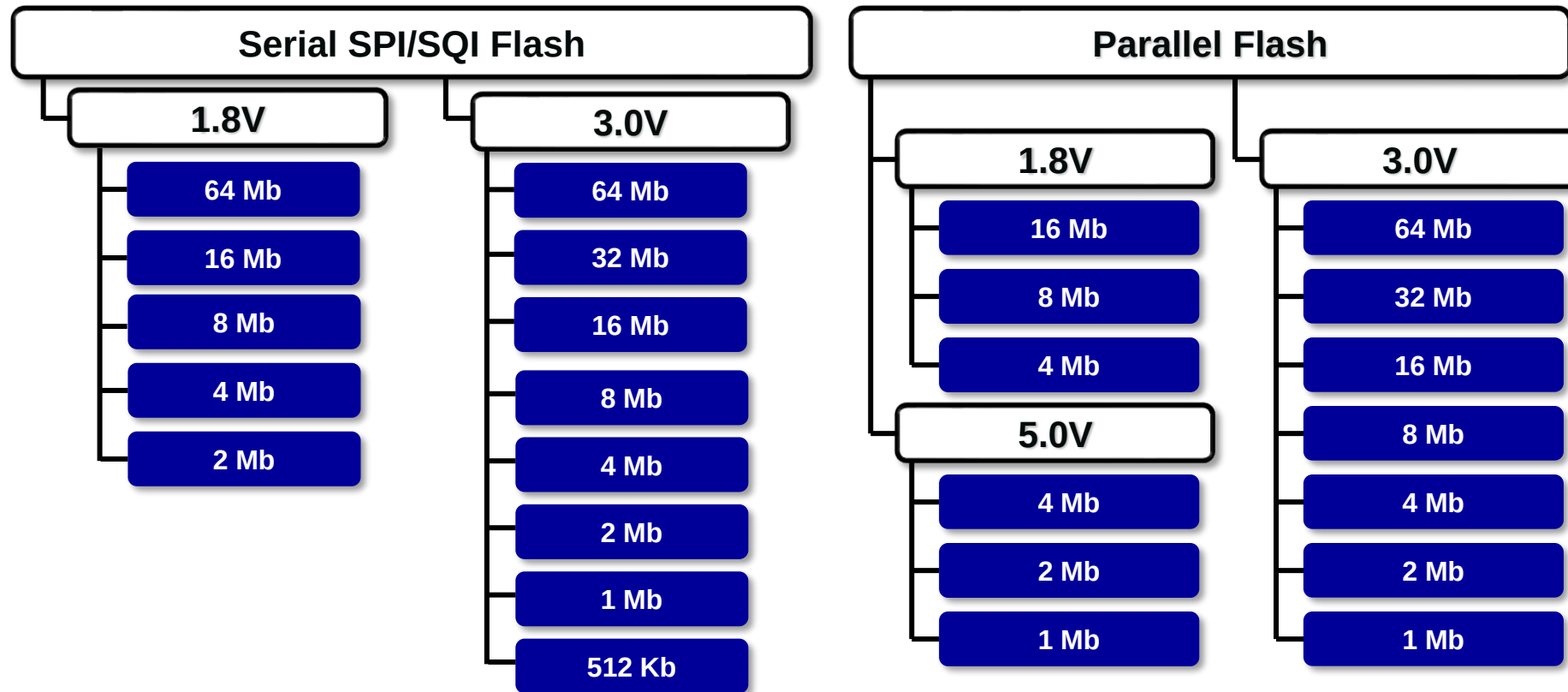
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4Mb SPI Serial EEPROM

	25CSM04
Organization	524,288 x 8
Operating Voltage	2.5 V to 5.5 V
Max Clock Freq	8 MHz
Page Size	256 Bytes
Write Protection	Quarter, Half, Whole Array or User-Definable
Temp Range	-40°C to 85°C
Endurance	1,000,000
Packages	SOIC-8, 8-Pad TDFN, 8-Ball CSP

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1.8V 2Mb Serial Flash

	SST25WF020A
Density	2Mb (x8)
Operating Voltage	1.65 V to 1.95 V
Bus	SPI
Max Frequency	40 MHz
Write Speed	3 ms (Page Program)
Page Size (bytes)	256
Standby Current	10 uA
Temp Range	-40°C to 85°C
Endurance	100,000 E/W cycles
Data Retention	20 years
Packages	SOIC-8, USON-8

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1.8V 4Mb Serial Flash

	SST25WF040B	SST26WF040B
Density	4Mb (x8)	4Mb (x8)
Operating Voltage	1.65 V to 1.95 V	1.65 V to 1.95 V
Bus	SPI	SPI, SDI, SQI
Max Frequency	40 MHz	104 MHz
Write Speed	0.8 ms / 256 bytes (Page Program)	1.5 ms (max) / 256 bytes (Page Program)
Page Size (bytes)	256	256
Standby Current	10 uA	10 uA
Temp Range	-40°C to 85°C	-40°C to 85°C
Endurance	100,000 E/W cycles	100,000 E/W cycles
Data Retention	20 years	100 years
Packages	SOIC-8, USON-8	SOIC-8, USON-8 (2x3mm), WDFN-8 (5x6 mm), CSP

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1.8V 8Mb Serial Flash

	SST25WF080B 	SST26WF080B
Density	8Mb (x8)	8Mb (x8)
Operating Voltage	1.65 V to 1.95 V	1.65 V to 1.95 V
Bus	SPI	SPI, SDI, SQI
Max Frequency	40 MHz	104 MHz
Write Speed	0.8 ms / 256 bytes (Page Program)	1.5 ms (max) / 256 bytes (Page Program)
Page Size (bytes)	256	256
Standby Current	10 uA	10 uA
Temp Range	-40°C to 85°C	-40°C to 85°C
Endurance	100,000 E/W cycles	100,000 E/W cycles
Data Retention	20 years	100 years
Packages	SOIC-8, USON-8	SOIC-8, USON-8 (2x3mm), WDFN-8 (5x6 mm), CSP

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1.8V 16Mb Serial Flash

	SST26WF016B
Density	16Mb (x8)
Operating Voltage	1.65 V to 1.95 V
Bus	SPI, SDI, SQI
Max Frequency	104 MHz
Write Speed	1.5 ms (max) / 256 bytes (Page Program)
Page Size (bytes)	256
Standby Current	10 uA
Temp Range	-40°C to 85°C
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	SOIC-8, WDFN-8 (5x6 mm), CSP

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1.8V 64Mb Serial Flash

	SST26WF064C
Density	64Mb (x8)
Operating Voltage	1.65 V to 1.95 V
Bus	SPI, SDI, SQI
Max Frequency	104 MHz
Write Speed	1.5 ms (max) / 256 bytes (Page Program)
Page Size (bytes)	256
Standby Current	10 uA
Temp Range	-40°C to 85°C
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	SOIC-8, WDFN-8 (5x6 mm), CSP

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3V 8Mb Serial Flash

	SST26VF080A 	SST25VF080B
Density	8Mb (x8)	8Mb (x8)
Operating Voltage	2.3 V to 3.6 V	2.7 V to 3.6 V
Bus	SQI, SPI	SPI
Max Frequency	104 MHz	50 MHz
Write Speed	1.5 ms (Page Program)	7 us (Word Program)
Standby Current	15 uA	5 uA
Temp Range	-40°C to 85°C (I-temp) -40°C to 125°C (E-temp)	0°C to 70°C (C-temp) -40°C to 85°C (I-temp)
Endurance	100,000 E/W cycles	100,000 E/W cycles
Data Retention	100 years	100 years
Packages	SOIC-8, WDFN-8 (5x6mm)	SOIC-8, SOIJ-8, PDIP-8, WSON-8 (5x6mm)

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3V 4Mb Serial Flash

	SST26VF040A 	SST25PF040C 	SST25VF040B
Density	4Mb (x8)	4Mb (x8)	4Mb (x8)
Operating Voltage	2.3 V to 3.6 V	2.3 V to 3.6 V	2.7 V to 3.6 V
Bus	SQI, SPI	SPI	SPI
Max Frequency	104 MHz	40 MHz	50 MHz
Write Speed	1.5 ms (Page Program)	4 ms / 256 bytes (Page Program)	7 us (Word Program)
Standby Current	15 uA	5 uA	5 uA
Temp Range	-40°C to 85°C (I-temp) -40°C to 125°C (E-temp)	-40°C to 85°C (I-temp) -40°C to 105°C (V-temp) -40°C to 125°C (E-temp)	0°C to 70°C (C-temp) -40°C to 85°C (I-temp)
Endurance	100,000 E/W cycles	100,000 E/W cycles	100,000 E/W cycles
Data Retention	100 years	20 years	100 years
Packages	SOIC-8, WDFN-8 (5x6mm)	SOIC-8, WDFN-8 (5x6mm), USON-8 (2x3mm)	SOIC-8, SOIJ-8, PDIP-8, WSON-8 (5x6mm), CSP

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3V 2Mb Serial Flash

	SST26VF020A 	SST25VF020	SST25xF020B	SST25LF020A
Density	4Mb (x8)	2Mb (x8)	2Mb (x8)	2Mb (x8)
Operating Voltage	2.3 V to 3.6 V	2.7 V to 3.6 V	2.3 V to 3.6 V	3 V to 3.6 V
Bus	SQI, SPI	SPI	SPI	SPI
Max Frequency	104 MHz	20 MHz	80 MHz	33 MHz
Write Speed	1.5 ms (Page Program)	14 us (Byte Program)	7 us (Byte Program)	14 us (Byte Program)
Standby Current	15 uA	8 uA	5 uA	8 uA
Temp Range	-40°C to 85°C (I-temp) -40°C to 125°C (E-temp)	0°C to 70°C (C-temp) -20°C to 85°C (E-temp) -40°C to 85°C (I-temp)	0°C to 70°C (C-temp) -40°C to 85°C (I-temp)	0°C to 70°C (C-temp) -20°C to 85°C (E-temp) -40°C to 85°C (I-temp)
Endurance	100,000 E/W cycles	100,000 E/W cycles	100,000 E/W cycles	100,000 E/W cycles
Data Retention	100 years	100 years	100 years	100 years
Packages	SOIC-8, WDFN-8 (5x6mm)	SOIC-8, WSON-8 (5x6mm)	SOIC-8, WSON-8 (5x6mm), USON-8 (2x3mm)	SOIC-8, WSON-8 (5x6mm)

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3V 1Mb Serial Flash

	SST25VF010A
Density	1Mb (x8)
Operating Voltage	2.7 V to 3.6 V
Bus	SPI
Max Frequency	33 MHz
Write Speed	14 us (Byte Program)
Standby Current	8 uA
Temp Range	0°C to 70°C (C-temp) -40°C to 85°C (I-temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	SOIC-8, WSON-8 (5x6mm), CSP

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3V 512Kb Serial Flash

	SST25VF512	SST25VF512A
Density	512Kb (x8)	512Kb (x8)
Operating Voltage	2.7 V to 3.6 V	2.7 V to 3.6 V
Bus	SPI	SPI
Max Frequency	20 MHz	33 MHz
Write Speed	14 us (Byte Program)	14 us (Byte Program)
Standby Current	8 uA	8 uA
Temp Range	0°C to 70°C	0°C to 70°C (C-temp) -40°C to 85°C (I-temp)
Endurance	100,000 E/W cycles	100,000 E/W cycles
Data Retention	100 years	100 years
Packages	SOIC-8	SOIC-8, WSON-8 CSP

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3V 32Mb Serial Flash

	SST26VF032B 
Density	32Mb (x8)
Operating Voltage	2.3 V to 3.6 V
Bus	SQI, SPI
Max Frequency	104 MHz
Write Speed	1.5 ms (Page Program)
Page Size (bytes)	256
Standby Current	15 uA
Temp Range	-40°C to 85°C (I-temp) -40°C to 105°C (V-temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	SOIC-8, WSON-8 (5x6mm), TFBGA-24 (6x8mm)

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3V 64Mb Serial Flash

	SST26VF064B 
Density	64Mb (x8)
Operating Voltage	2.3 V to 3.6 V
Bus	SQI, SPI, SDI
Max Frequency	104 MHz
Write Speed	1.5 ms (Page Program)
Page Size (bytes)	256
Standby Current	15 uA
Temp Range	-40°C to 85°C (I-temp) -40°C to 105°C (V-temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	SOIC-8, SOIC-16, WSON-8 (5x6mm), TFBG-24 (6x8mm)

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3V 16Mb Serial Flash

	SST25VF016B	SST26VF016B 
Density	16Mb (x8)	16Mb (x8)
Operating Voltage	2.7 V to 3.6 V	2.3 V to 3.6 V
Bus	SPI	SQI, SPI
Max Frequency	75 MHz	104 MHz
Write Speed	7 us (Word Program)	1.5 ms (Page Program)
Page Size (bytes)	5 uA	256
Standby Current	0°C to 70°C (C-temp) -40°C to 85°C (I-temp)	15 uA
Temp Range	100,000 E/W cycles	-40°C to 85°C (I-temp) -40°C to 105°C (V-temp) -40°C to 125°C (E-temp)
Endurance	100 years	100,000 E/W cycles
Data Retention	SOIC-8, WSON-8 (6x5mm)	100 years
Packages	16Mb (x8)	SOIC-8, WSON-8 (5x6mm), TFBGA-24 (6x8mm)

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1.8V 16Mb Parallel Flash

	SST39WF1601/2 
Density	16Mb (x16)
Operating Voltage	1.65 V to 1.95 V
Bus	Parallel
Read Access Time	70 ns
Write Speed	28 us (Word Program)
Standby Current	5 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TFBGA-48 (6x8mm), WFBGA-48 (4x6mm)
Block Protection	Top (SST39WF1602) or Bottom (SST39WF1601)

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1.8V 8Mb Parallel Flash

	SST39WF800B 
Density	8Mb (x16)
Operating Voltage	1.65 V to 1.95 V
Bus	Parallel
Read Access Time	70 ns
Write Speed	28 us (Word Program)
Standby Current	5 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TFBGA-48 (6x8mm), WFBGA-48 (4x6mm)

[Online
Datasheet](#)

1.8V 4Mb Parallel Flash

	SST39WF400B 
Density	4Mb (x16)
Operating Voltage	1.65 V to 1.95 V
Bus	Parallel
Read Access Time	70 ns
Write Speed	28 us (Word Program)
Standby Current	5 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TFBGA-48 (6x8mm), WFBGA-48 (4x6mm)

[Online
Datasheet](#)

5V 4Mb Parallel Flash

	SST39SF040
Density	4Mb (x8)
Operating Voltage	4.5 V to 5.5 V
Bus	Parallel
Read Access Time	55 ns, 70 ns
Write Speed	14 us (Byte Program)
Standby Current	30 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TSOP-32 (8x14mm), PLCC-32, PDIP-32

[Online
Datasheet](#)

5V 2Mb Parallel Flash

	SST39SF020A
Density	2Mb (x8)
Operating Voltage	4.5 V to 5.5 V
Bus	Parallel
Read Access Time	55 ns, 70 ns
Write Speed	14 us (Byte Program)
Standby Current	30 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TSOP-32 (8x14mm), PLCC-32, PDIP-32

[Online
Datasheet](#)

5V 1Mb Parallel Flash

	SST39SF010A
Density	1Mb (x8)
Operating Voltage	4.5 V to 5.5 V
Bus	Parallel
Read Access Time	55 ns, 70 ns
Write Speed	14 us (Byte Program)
Standby Current	30 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TSOP-32 (8x14mm), PLCC-32, PDIP-32

[Online
Datasheet](#)

3V 64Mb Parallel Flash

	SST39VF640xB 	SST38VF640x
Density	64Mb (x16)	64Mb (x16)
Operating Voltage	2.7 V to 3.6 V	2.7 V to 3.6 V
Bus	Parallel	Parallel
Read Access Time	70 ns	70 ns
Write Speed	7 us (Word Program)	7 us (Word Program)
Standby Current	3 uA	3 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles	100,000 E/W cycles
Data Retention	100 years	100 years
Packages	TSOP-48, TFBGA-48	TSOP-48, TFBGA-48
Block Protection	Top (SST39VF6402B) or Bottom (SST39VF6401B)	Top (SST38VF6402/4) or Bottom (SST38VF6401/3)
Boot Block Erase	Uniform	Uniform 32 KWord (SST38VF6401/2) Non-Uniform 8 Kword (SST38VF6403/4)

Online
Datasheet

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3V 32Mb Parallel Flash

	SST39VF320xC 
Density	32Mb (x16)
Operating Voltage	2.7 V to 3.6 V
Bus	Parallel
Read Access Time	70 ns
Write Speed	7 us (Word Program)
Standby Current	4 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TSOP-48 (12x20mm), TFBGA-48 (6x8mm)
Block Protection	Top (SST39VF3202C) or Bottom (SST39VF3201C)

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Datasheet



= Recommended for Automotive Designs

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3V 16Mb Parallel Flash

	SST39VF160xC
Density	16Mb (x16)
Operating Voltage	2.7 V to 3.6 V
Bus	Parallel
Read Access Time	70 ns
Write Speed	7 us (Word Program)
Standby Current	3 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TSOP-48 (12x20mm), TFBGA-48 (6x8mm), WFBGA-48 (4x6mm)
Block Protection	Top (SST39VF1602C) or Bottom (SST39VF1601C)

[Online
Datasheet](#)

3V 8Mb Parallel Flash

	SST39xF80xC
Density	8Mb (x16)
Operating Voltage	3.0 V to 3.6 V (SST39LF80xC), 2.7 V to 3.6 V (SST39VF80xC)
Bus	Parallel
Read Access Time	55 ns (SST39LF80xC), 70 ns (SST39VF80xC)
Write Speed	7 μ s (Word Program)
Standby Current	3 μ A
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TSOP-48 (12x20mm), TFBGA-48 (6x8mm), WFBGA-48 (4x6mm)
Block Protection	Top (SST39VF802C) or Bottom (SST39VF801C)

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Datasheet](#)

3V 4Mb Parallel Flash

	SST39xF40xC	SST39VF040
Density	4Mb (x16)	4Mb (x8)
Operating Voltage	3.0 V to 3.6 V (SST39LF40xC) 2.7 V to 3.6 V (SST39VF40xC)	2.7 V to 3.6 V
Bus	Parallel	Parallel
Read Access Time	55 ns (SST39LF40xC) 70 ns (SST39VF40xC)	70 ns
Write Speed	7 us (Word Program)	14 us (Byte Program)
Standby Current	3 uA	1 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles	100,000 E/W cycles
Data Retention	100 years	100 years
Packages	TSOP-48 (12x20mm), TFBGA-48 (6x8mm), WFBGA-48 (4x6mm)	TSOP-32 (8x14mm), PLCC-32

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Datasheet](#)

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3V 2Mb Parallel Flash

	SST39VF020	SST39VF200A
Density	2Mb (x8)	2Mb (x16)
Operating Voltage	2.7 V to 3.6 V	2.7 V to 3.6 V
Bus	Parallel	Parallel
Read Access Time	70 ns	70 ns
Write Speed	14 us (Byte Program)	14 us (Word Program)
Standby Current	1 uA	3 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles	100,000 E/W cycles
Data Retention	100 years	100 years
Packages	TSOP-32 (8x14mm), PLCC-32	TSOP-48 (12x20mm), TFBGA-48 (6x8mm), WFBGA-48 (4x6mm)

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Datasheet](#)

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3V 1Mb Parallel Flash

	SST39VF010
Density	1Mb (x8)
Operating Voltage	2.7 V to 3.6 V
Bus	Parallel
Read Access Time	70 ns
Write Speed	14 us (Byte Program)
Standby Current	1 uA
Temp Range	0°C to 70°C (C-Temp) -40°C to 85°C (I-Temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	PLCC-32, TSOP-32(8x14mm)

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Datasheet](#)

Intel Firmware Flash

Intel 82802 Firmware Hub (FWH)

SST49LF008A

Intel Low Pin Count (LPC)
Interface Specification 1.1

SST49LF080A

LPC Firmware Flash/Firmware Hub is a family of flash memories designed for system code and data storage. SST LPC (Low Pin Count) Flash and Firmware Hub Flash complies with the Intel LPC Interface specification supporting multi-byte Firmware Memory and single-byte LPC Memory cycle types.

Intel Firmware Hub

	SST49LF008A
Density	8Mb (x8)
Operating Voltage	3.0 V to 3.6 V
Bus	Intel Firmware Hub (FWH)
Max Frequency	33 MHz
Write Speed	14 us (Byte Program)
Standby Current (Max)	10 uA
Temp Range	0°C to 70°C
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TSOP-32, PLCC-32, TSOP-40

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Intel LPC Interface Specification

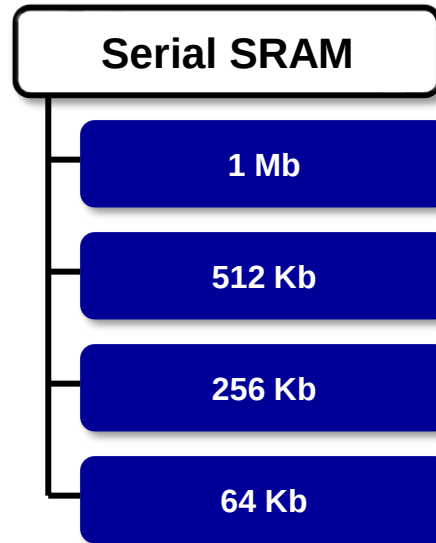
	SST49LF080A
Density	8Mb (x8)
Operating Voltage	3.0 V to 3.6 V
Bus	Intel Firmware Hub (FWH)
Max Frequency	33 MHz
Write Speed	14 us (Byte Program)
Standby Current (Max)	10 uA
Temp Range	0°C to 70°C
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	TSOP-32, PLCC-32

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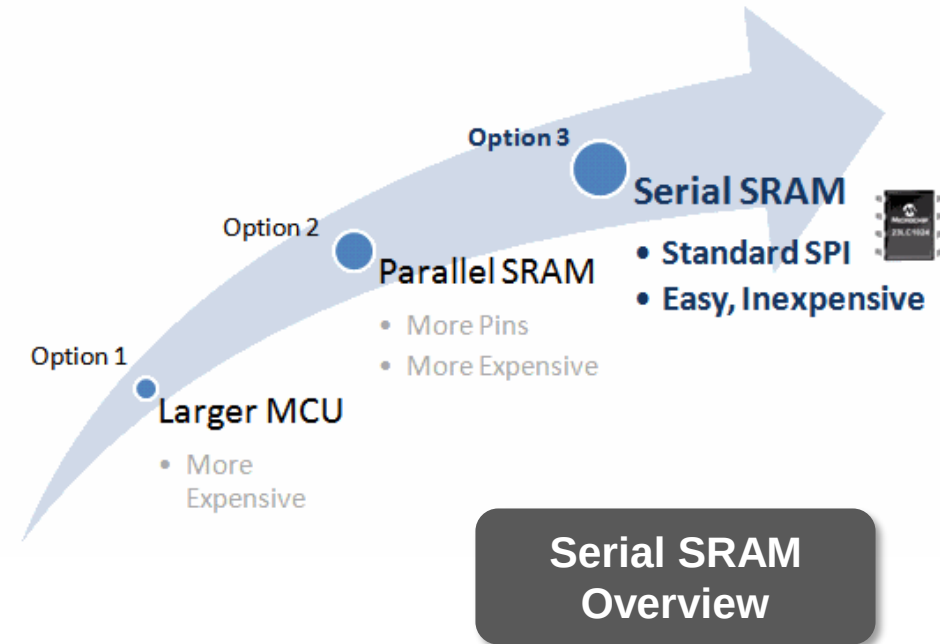
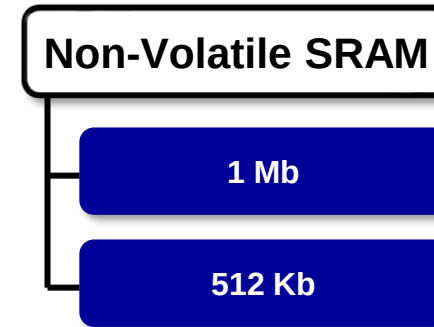
Serial SRAM

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Design &
Development Tools



- Supports SPI, SDI, & SQI bus modes
- Unlimited erase/write cycles
- Fast writes with zero latency
- Low Power Consumption
- Low Cost
- Small Footprint package



1Mb Serial SRAM

	23x1024 
Density	1Mb (x8)
Operating Voltage	1.7 V to 2.2 V (23A1024) 2.5 V to 5.5 V (23LC1024)
Bus	SPI / SDI / SQI
Max Frequency	20 MHz
Page Size (bytes)	32
Temp Range	-40°C to 85°C -40°C to 125°C (E-temp only)
Endurance	Unlimited
Packages	SOIC-8, PDIP-8, TSSOP-8

[Online
Datasheet](#)

512Kb Serial SRAM

	23x512 
Density	512Kb (x8)
Operating Voltage	1.7 V to 2.2 V (23A512) 2.5 V to 5.5 V (23LC512)
Bus	SPI / SDI / SQI
Max Frequency	20 MHz
Page Size (bytes)	32
Temp Range	-40°C to 85°C -40°C to 125°C (E-temp only)
Endurance	Unlimited
Packages	SOIC-8, PDIP-8, TSSOP-8

[Online
Datasheet](#)

256Kb Serial SRAM

	23x256 
Density	256Kb (x8)
Operating Voltage	1.7 V to 1.95 V (23A256) 2.7 V to 3.6 V (23K256)
Bus	SPI
Max Frequency	20 MHz
Page Size (bytes)	32
Temp Range	-40°C to 85°C -40°C to 125°C (E-temp only)
Endurance	Unlimited
Packages	SOIC-8, PDIP-8, TSSOP-8

[Online Datasheet](#)

64Kb Serial SRAM

	23x640 
Density	64Kb (x8)
Operating Voltage	1.5 V to 1.95 V (23A640) 2.7 V to 3.6 V (23K640)
Bus	SPI
Max Frequency	20 MHz
Page Size (bytes)	32
Temp Range	-40°C to 85°C -40°C to 125°C (E-temp only)
Endurance	Unlimited
Packages	SOIC-8, PDIP-8, TSSOP-8

[Online Datasheet](#)

1Mb Serial NVSRAM

	23LCV1024 
Density	1Mb (x8)
Operating Voltage	2.5 V to 5.5 V
Bus	SPI / SDI
Max Frequency	20 MHz
Page Size (bytes)	32
Temp Range	-40°C to 85°C
Endurance	Unlimited
Battery Backup	Yes
Packages	SOIC-8, PDIP-8, TSSOP-8

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Datasheet](#)

512Kb Serial NVSRAM

	23LCV512 
Density	512Kb (x8)
Operating Voltage	2.5 V to 5.5 V
Bus	SPI / SDI
Max Frequency	20 MHz
Page Size (bytes)	32
Temp Range	-40°C to 85°C
Endurance	Unlimited
Battery Backup	Yes
Packages	SOIC-8, PDIP-8, TSSOP-8

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Serial EERAM

[<< BACK](#)[Design & Development Tools](#)

EERAM combines the reliability of an EEPROM with the performance of an SRAM. It is a nonvolatile SRAM with a shadow EEPROM backup.

I²C Serial EERAM

16Kb

4 Kb

SPI Serial EERAM

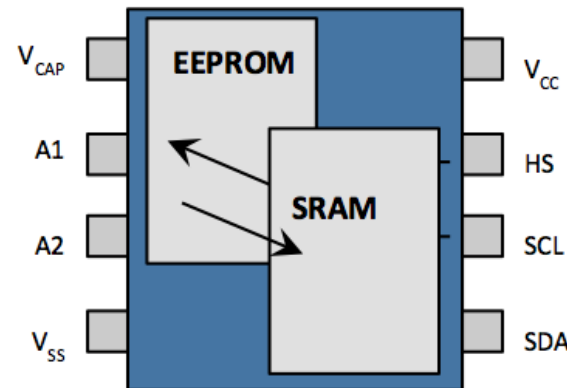
1 Mb

512 Kb

256 Kb

64 Kb

- 3.0V (SPI, I2C); 5.0V (I2C)
- Nonvolatile (with external capacitor)
- Unlimited SRAM erase/write cycles
- Instantaneous, random read/write



Serial EERAM
Overview

4Kb I²C Serial EERAM

	47x04 
Organization	512 x 8 (4Kb)
Operating Voltage	2.7V to 3.6V (47L04) 4.5V to 5.5V (47C04)
Bus	I ² C
Max Frequency	1 MHz
Write Protection	Software from 1/64 of SRAM array to whole array
Temp Range	-40°C to 85°C -40°C to 125°C (E-temp only)
Endurance	Unlimited (SRAM); 1,000,000 (EEPROM)
Data Retention	200+ Years
Packages	SOIC-8, PDIP-8, TSSOP-8

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Datasheet](#)

16Kb I²C Serial EERAM

	47x16 
Organization	2048 x 8 (16Kb)
Operating Voltage	2.7V to 3.6V (47L16) 4.5V to 5.5V (47C16)
Bus	I ² C
Max Frequency	1 MHz
Write Protection	Software from 1/64 of SRAM array to whole array
Temp Range	-40°C to 85°C -40°C to 125°C (E-temp only)
Endurance	Unlimited (SRAM); 1,000,000 (EEPROM)
Data Retention	200+ Years
Packages	SOIC-8, PDIP-8, TSSOP-8

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1 Mb SPI Serial EERAM

	48LM01
Organization	131,072 x 8 (1Mb)
Operating Voltage	2.7V to 3.6V
Bus	SPI
Max Frequency	66 MHz
Write Protection	Software ¼, ½, or Full Array
Temp Range	-40°C to 85°C
Endurance	Unlimited (SRAM); 100,000 (EEPROM)
Data Retention	100+ Years
Packages	SOIJ-8

[Online
Datasheet](#)

512 Kb SPI Serial EERAM

	48L512
Organization	65,536 x 8 (512Kb)
Operating Voltage	2.7V to 3.6V
Bus	SPI
Max Frequency	66 MHz
Write Protection	Software ¼, ½, or Full Array
Temp Range	-40°C to 85°C
Endurance	Unlimited (SRAM); 100,000 (EEPROM)
Data Retention	100+ Years
Packages	SOIC-8

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Datasheet](#)

256 Kb SPI Serial EERAM

	48L256
Organization	32,768 x 8 (256Kb)
Operating Voltage	2.7V to 3.6V
Bus	SPI
Max Frequency	66 MHz
Write Protection	Software ¼, ½, or Full Array
Temp Range	-40°C to 85°C
Endurance	Unlimited (SRAM); 100,000 (EEPROM)
Data Retention	100+ Years
Packages	SOIC-8

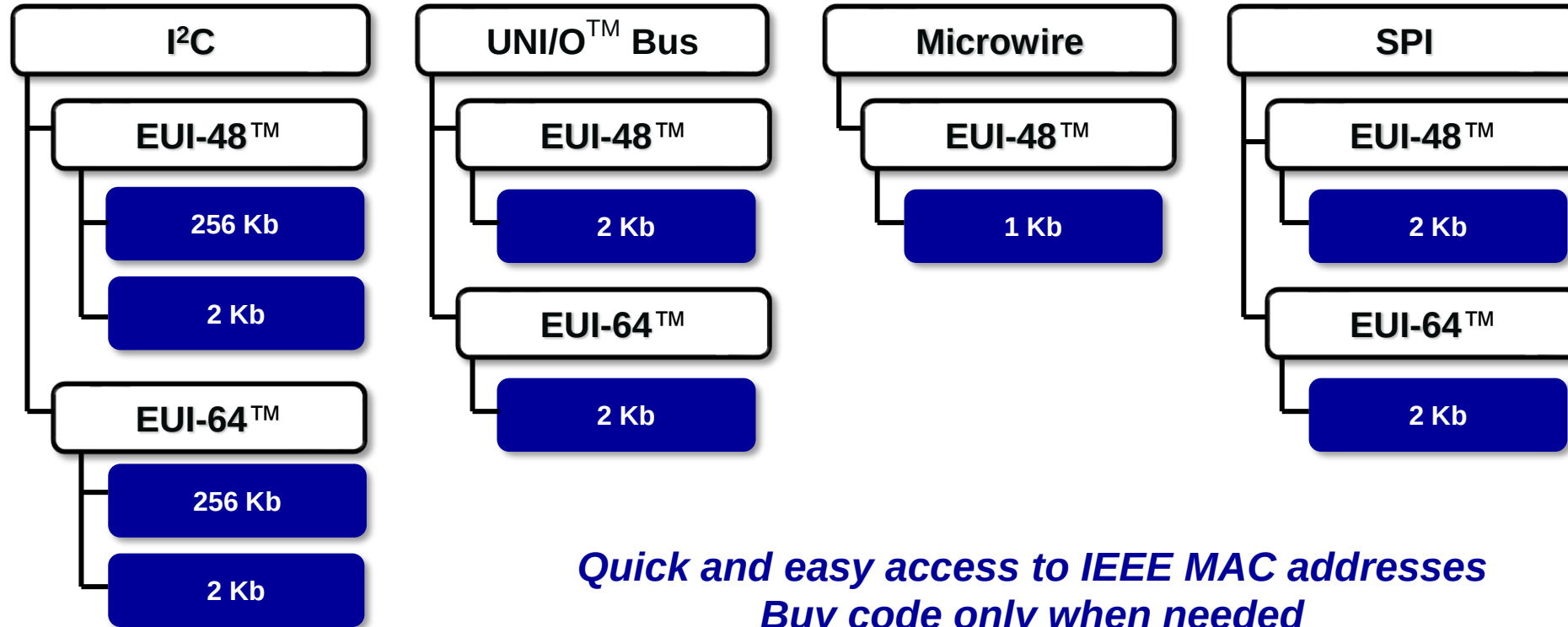
[Online
Datasheet](#)

64 Kb SPI Serial EERAM

	48L640
Organization	8,192 x 8 (64Kb)
Operating Voltage	2.7V to 3.6V
Bus	SPI
Max Frequency	66 MHz
Write Protection	Software ¼, ½, or Full Array
Temp Range	-40°C to 85°C
Endurance	Unlimited (SRAM); 100,000 (EEPROM)
Data Retention	100+ Years
Packages	SOIC-8, TDFN-8

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Datasheet](#)

Pre-Programmed MAC Address EEPROMs



Quick and easy access to IEEE MAC addresses
Buy code only when needed
No added programming and serialization cost
No volume restrictions

- EUI-48: Networking, Ethernet, Wi-Fi (IEEE 802.11), Bluetooth
- EUI-64: ZigBee (IEEE 802.15.4), MiWi, FireWire, IPv6

2Kb I²C w/ EUI-48™

	24AA02E48	24AA025E48	AT24MAC402
Density	2 Kb (x8)	2 Kb (x8)	2 Kb (x8)
ID Type	EUI-48	EUI-48	EUI-48
Operating Voltage	1.7 V to 5.5 V	1.7 V to 5.5 V	1.7 V to 5.5 V
Max Clock Freq	400 KHz	400 KHz	1 MHz
Page Size	8 Bytes	16 Bytes	16 Bytes
Write Protection	Half Array	Half Array	Whole Array
Temp Range	-40°C to +85°C	-40°C to +85°C	-40°C to +85°C
Endurance	1,000,000	1,000,000	1,000,000
Data Retention	200 years	200 years	100 years
Packages	SOT-23-5, SOIC-8	SOT-23-6, SOIC-8	SOT-23-5, SOIC-8, TSSOP, UDFN

[Online Datasheet](#)

[Online Datasheet](#)

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256Kb I²C w/ EUI-48/64™

	24AA256UID
Density	256 Kb (x8)
ID Type	EUI-48, EUI-64, and Unique 32-bit serial number
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	64 Bytes
Write Protection	Quarter Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	TSSOP-8, SOIC-8, PDIP-8

[Online Datasheet](#)

2Kb I²C w/ EUI-64™

	24AA02E48	24AA025E48	AT24MAC602
Density	2 Kb (x8)	2 Kb (x8)	2 Kb (x8)
ID Type	EUI-64	EUI-64	EUI-64
Operating Voltage	1.7 V to 5.5 V	1.7 V to 5.5 V	1.7 V to 5.5 V
Max Clock Freq	400 KHz	400 KHz	1 MHz
Page Size	8 Bytes	16 Bytes	16 Bytes
Write Protection	Half Array	Half Array	Whole Array
Temp Range	-40°C to +85°C	-40°C to +85°C	-40°C to +85°C
Endurance	1,000,000	1,000,000	1,000,000
Data Retention	200 years	200 years	100 years
Packages	SOT-23-5, SOIC-8	SOT-23-6, SOIC-8	SOT-23-5, SOIC-8, TSSOP, UDFN

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[Online Datasheet](#)

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2Kb UNI/O[®] w/ EUI-48™

	11AA02E48
Density	2 Kb (x8)
ID Type	EUI-48
Bus Type	UNI/O [®]
Operating Voltage	1.8 V to 5.5 V
Max Clock Freq	100 KHz
Page Size	16 Bytes
Write Protection	Quarter Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	SOT-23-3, SOIC-8

[Online
Datasheet](#)

2Kb UNI/O[®] w/ EUI-64[™]

	11AA02E64
Density	2 Kb (x8)
ID Type	EUI-64
Bus Type	UNI/O [®]
Operating Voltage	1.8 V to 5.5 V
Max Clock Freq	100 KHz
Page Size	16 Bytes
Write Protection	Quarter Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	SOT-23-3, SOIC-8

[Online
Datasheet](#)

2Kb SPI w/ EUI-48™

	25AA02E48
Density	2 Kb (x8)
ID Type	EUI-48
Bus Type	SPI
Operating Voltage	1.8 V to 5.5 V
Max Clock Freq	10 MHz
Page Size	16 Bytes
Write Protection	Quarter Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	SOT-23-6, SOIC-8

[Online
Datasheet](#)

2Kb SPI w/ EUI-64™

	25AA02E64
Density	2 Kb (x8)
ID Type	EUI-64
Bus Type	SPI
Operating Voltage	1.8 V to 5.5 V
Max Clock Freq	10 MHz
Page Size	16 Bytes
Write Protection	Quarter Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	SOT-23-6, SOIC-8

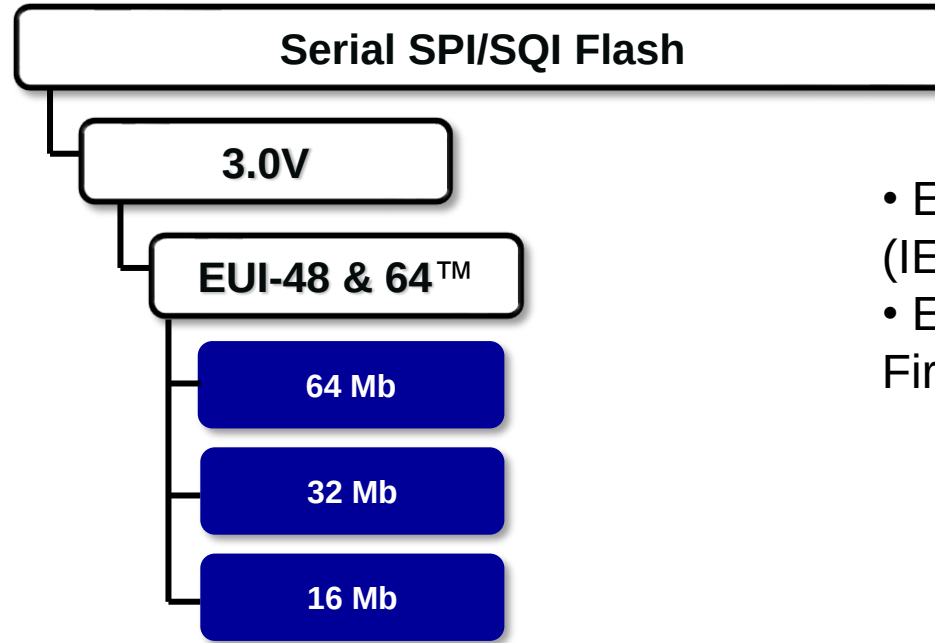
[Online
Datasheet](#)

2Kb Microwire w/ EUI-48™

	93AA46AE48
Density	2 Kb (x8)
ID Type	EUI-48
Bus Type	Microwire
Operating Voltage	1.8 V to 5.5 V
Max Clock Freq	2 MHz
Page Size	16 Bytes
Write Protection	None
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	SOT-23-6, SOIC-8

[Online
Datasheet](#)

Pre-Programmed MAC Address NOR Flash



- EUI-48: Networking, Ethernet, Wi-Fi (IEEE 802.11), Bluetooth
- EUI-64: ZigBee (IEEE 802.15.4), MiWi, FireWire, IPv6

Quick and easy access to IEEE MAC addresses

Buy code only when needed

No added programming and serialization cost

No volume restrictions

*Contact Marketing for temperature and package options not listed

3V 16Mb Serial Flash w/ EUI-48/64™

	SST26VF016BEUI 
Density	16Mb (x8)
ID Type	EUI-48 and EUI-64
Operating Voltage	2.3 V to 3.6 V
Bus	SQI, SPI
Max Frequency	104 MHz
Write Speed	1.5 ms (Page Program)
Page Size (bytes)	256
Standby Current	15 uA
Temp Range	-40°C to 85°C (I-temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	SOIC-8

[Online Datasheet](#)

3V 32Mb Serial Flash w/ EUI-48/64™

	SST26VF032BEUI 
Density	32Mb (x8)
ID Type	EUI-48 and EUI-64
Operating Voltage	2.3 V to 3.6 V
Bus	SQI, SPI
Max Frequency	104 MHz
Write Speed	1.5 ms (Page Program)
Page Size (bytes)	256
Standby Current	15 uA
Temp Range	-40°C to 85°C (I-temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	SOIC-8

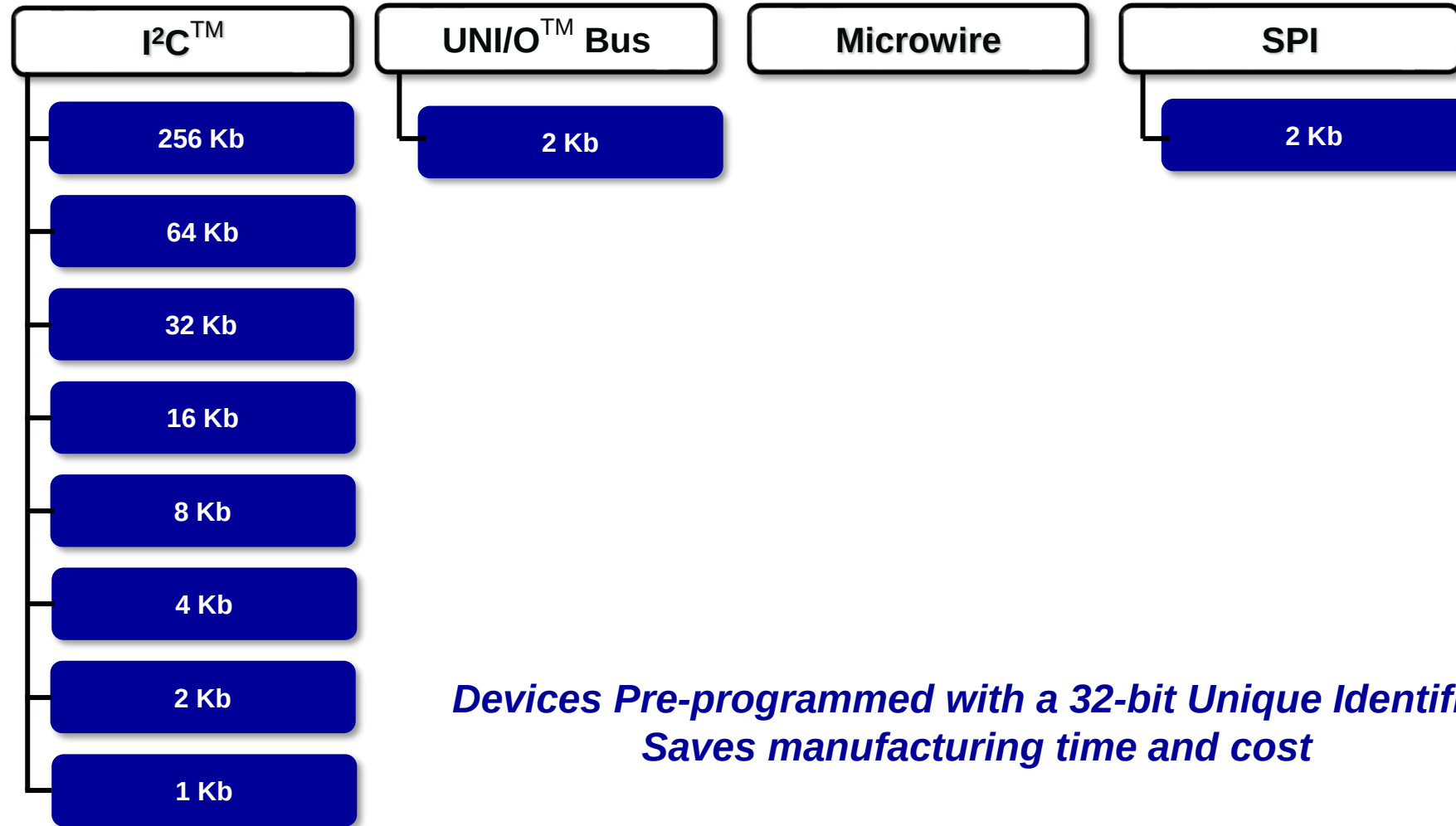
[Online Datasheet](#)

3V 64Mb Serial Flash w/ EUI-48/64™

	SST26VF064BEUI 
Density	64Mb (x8)
ID Type	EUI-48 and EUI-64
Operating Voltage	2.3 V to 3.6 V
Bus	SQI, SPI, SDI
Max Frequency	104 MHz
Write Speed	1.5 ms (Page Program)
Page Size (bytes)	256
Standby Current	15 uA
Temp Range	-40°C to 85°C (I-temp)
Endurance	100,000 E/W cycles
Data Retention	100 years
Packages	SOIC-8, WSON-8 (5x6mm)

[Online Datasheet](#)

Unique ID



*Devices Pre-programmed with a 32-bit Unique Identifier
Saves manufacturing time and cost*

1Kb I²C w/ Unique ID

	AT24CS01
Density	1 Kb (x8)
ID Type (Serial #)	Unique 128-bit
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	8 Bytes
Write Protection	Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

[Online
Datasheet](#)

2Kb I²C w/ Unique ID

	24AA02UID	24AA025UID	AT24CS02
Density	2 Kb (x8)	2 Kb (x8)	2 Kb (x8)
ID Type (Serial #)	Unique 32-bit	Unique 32-bit	Unique 128-bit
Operating Voltage	1.7 V to 5.5 V	1.7 V to 5.5 V	1.7 V to 5.5 V
Max Clock Freq	400 KHz	400 KHz	1 MHz
Page Size	8 Bytes	16 Bytes	8 Bytes
Write Protection	Half Array	Half Array	Whole Array
Temp Range	-40°C to +85°C	-40°C to +85°C	-40°C to +85°C
Endurance	1,000,000	1,000,000	1,000,000
Data Retention	200 years	200 years	100 years
Packages	SOT-23-5, SOIC-8, PDIP-8	SOT-23-6, SOIC-8, PDIP-8	SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

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Datasheet](#)

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<< BACK

4Kb I²C w/ Unique ID

	AT24CS04
Density	4 Kb (x8)
ID Type (Serial #)	Unique 128-bit
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	16 Bytes
Write Protection	Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

[Online Datasheet](#)

8Kb I²C w/ Unique ID

	AT24CS08
Density	8 Kb (x8)
ID Type (Serial #)	Unique 128-bit
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	16 Bytes
Write Protection	Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

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16Kb I²C w/ Unique ID

	AT24CS16
Density	16 Kb (x8)
ID Type (Serial #)	Unique 128-bit
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	16 Bytes
Write Protection	Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

[Online
Datasheet](#)

32Kb I²C w/ Unique ID

	AT24CS32
Density	32 Kb (x8)
ID Type (Serial #)	Unique 128-bit
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	32 Bytes
Write Protection	Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

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Datasheet](#)

64Kb I²C w/ Unique ID

	AT24CS64
Density	64 Kb (x8)
ID Type (Serial #)	Unique 128-bit
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	32 Bytes
Write Protection	Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	SOIC-8, TSSOP-8, UDFN-8

[Online Datasheet](#)

256Kb I²C w/ Unique ID

	24AA256UID
Density	256 Kb (x8)
ID Type (Serial #)	Unique 32-bit serial number and EUI-48 or EUI-64
Operating Voltage	1.7 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	64 Bytes
Write Protection	Quarter Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	TSSOP-8, SOIC-8, PDIP-8

[Online Datasheet](#)

2Kb UNI/O® w/ Unique ID

	11AA02UID
Density	2 Kb (x8)
ID Type (Serial #)	Unique 32-bit serial number
Operating Voltage	1.8 V to 5.5 V
Max Clock Freq	100 KHz
Page Size	16 Bytes
Write Protection	Quarter Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	SOT-23-3, SOIC-8, PDIP-8

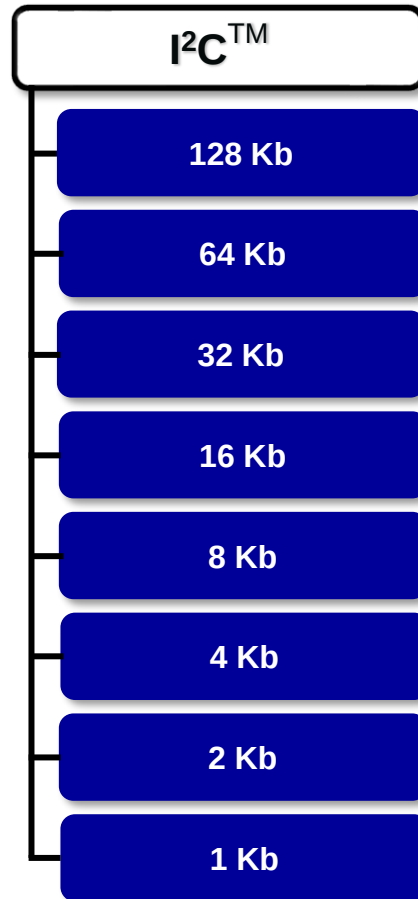
[Online Datasheet](#)

2Kb SPI w/ Unique ID

	25AA02UID
Density	2 Kb (x8)
ID Type (Serial #)	Unique 32-bit serial number
Operating Voltage	1.8 V to 5.5 V
Max Clock Freq	10 MHz
Page Size	16 Bytes
Write Protection	Quarter Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	SOT-23-6, SOIC-8, PDIP-8

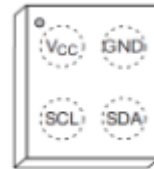
[Online Datasheet](#)

4-Ball WLCSP



Feature small size, thin profile, and low power consumption

- Replaces HW write protection with 4-zone lockable software write protection
- Removes Address Pins -> hardware address configurable from factory



Application Example: Camera Module

- Store parameters on brightness, contrast, resolution, compression ratio, etc..
- Position info for Image Stabilization (OIS) and Autofocusing (AF)
- Factory settings, firmware, and serial numbers



1Kb I²C 4-Ball WLCSP

	AT24CSW010
Density	1 Kb (x8)
ID Type	256-bit security register w/ pre-programmed unique 128-bit serial #
Operating Voltage	1.7 V to 3.6 V
Max Clock Freq	1 MHz
Page Size	16 Bytes
Write Protection	Quarter, Half or Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	WLCSP

[Online
Datasheet](#)

2Kb I²C 4-Ball WLCSP

	AT24CSW020
Density	2 Kb (x8)
ID Type	256-bit security register w/ pre-programmed unique 128-bit serial #
Operating Voltage	1.7 V to 3.6 V
Max Clock Freq	1 MHz
Page Size	16 Bytes
Write Protection	Quarter, Half or Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	WLCSP

[Online
Datasheet](#)

4Kb I²C 4-Ball WLCSP

	AT24CSW040
Density	4 Kb (x8)
ID Type	256-bit security register w/ pre-programmed unique 128-bit serial #
Operating Voltage	1.7 V to 3.6 V
Max Clock Freq	1 MHz
Page Size	16 Bytes
Write Protection	Quarter, Half or Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	WLCSP

[Online
Datasheet](#)

8Kb I²C 4-Ball WLCSP

	AT24CSW080
Density	8 Kb (x8)
ID Type	256-bit security register w/ pre-programmed unique 128-bit serial #
Operating Voltage	1.7 V to 3.6 V
Max Clock Freq	1 MHz
Page Size	16 Bytes
Write Protection	Quarter, Half or Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	100 years
Packages	WLCSP

[Online
Datasheet](#)

16Kb I²C 4-Ball WLCSP

	AT24CW160
Density	16 Kb (x8)
ID Type	Programmable Hardware Slave Address Bits
Operating Voltage	1.6 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	32 Bytes
Write Protection	1/4, 1/2, 3/4 or Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	WLCSP, SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

[Online
Datasheet](#)

32Kb I²C 4-Ball WLCSP

	AT24CW320
Density	32 Kb (x8)
ID Type	Programmable Hardware Slave Address Bits
Operating Voltage	1.6 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	32 Bytes
Write Protection	1/4, 1/2, 3/4 or Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	WLCSP, SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

[Online
Datasheet](#)

64Kb I²C 4-Ball WLCSP

	24CW640
Density	64 Kb (x8)
ID Type	Programmable Hardware Slave Address Bits
Operating Voltage	1.6 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	32 Bytes
Write Protection	1/4, 1/2, 3/4 or Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	WLCSP, SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

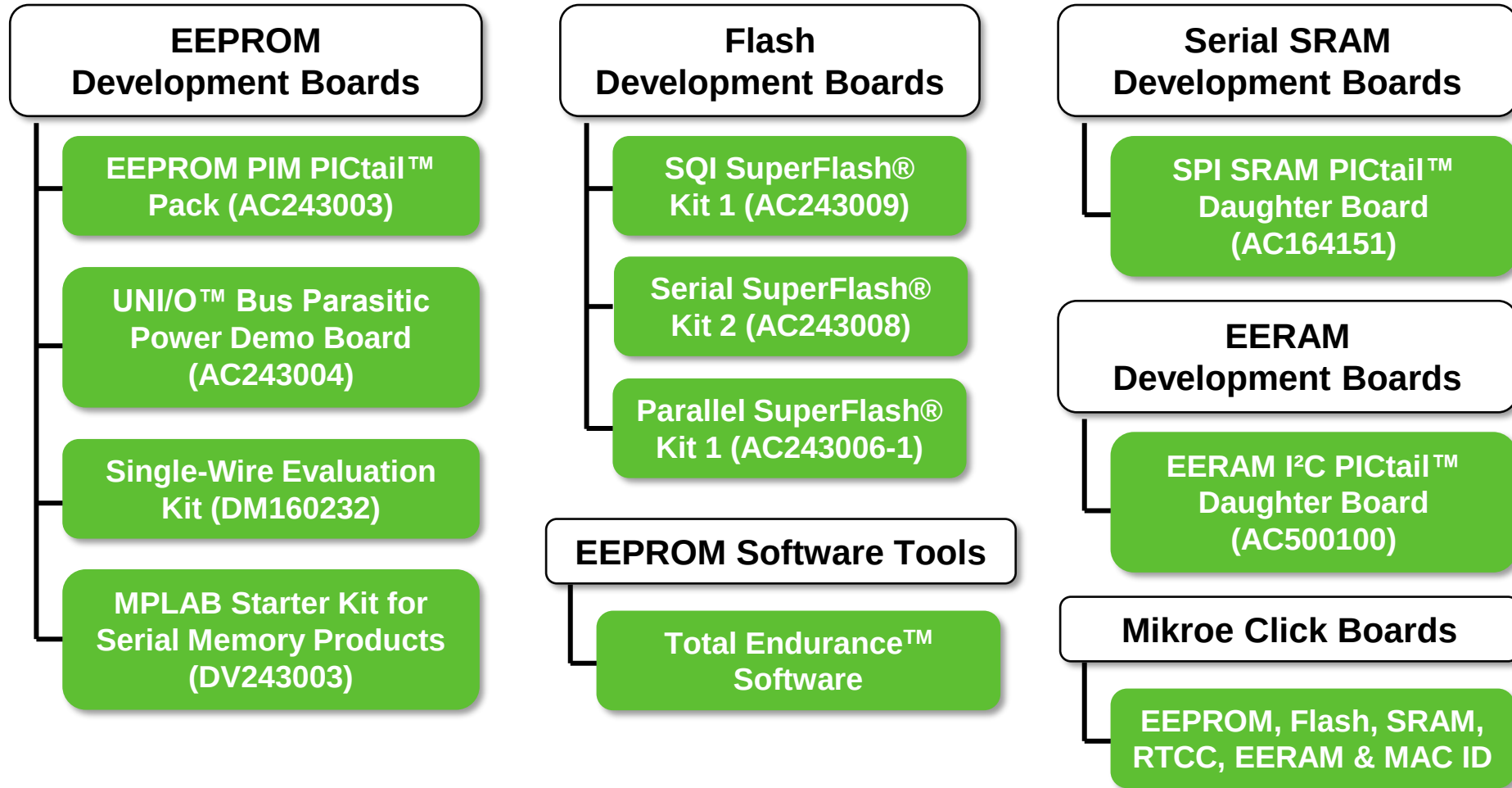
[Online
Datasheet](#)

128Kb I²C 4-Ball WLCSP

	24CW1280
Density	128 Kb (x8)
ID Type	Programmable Hardware Slave Address Bits
Operating Voltage	1.6 V to 5.5 V
Max Clock Freq	1 MHz
Page Size	32 Bytes
Write Protection	1/4, 1/2, 3/4 or Whole Array
Temp Range	-40°C to +85°C
Endurance	1,000,000
Data Retention	200 years
Packages	WLCSP, SOT-23-5, SOIC-8, TSSOP-8, UDFN-8

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Datasheet](#)

Design & Development Tools



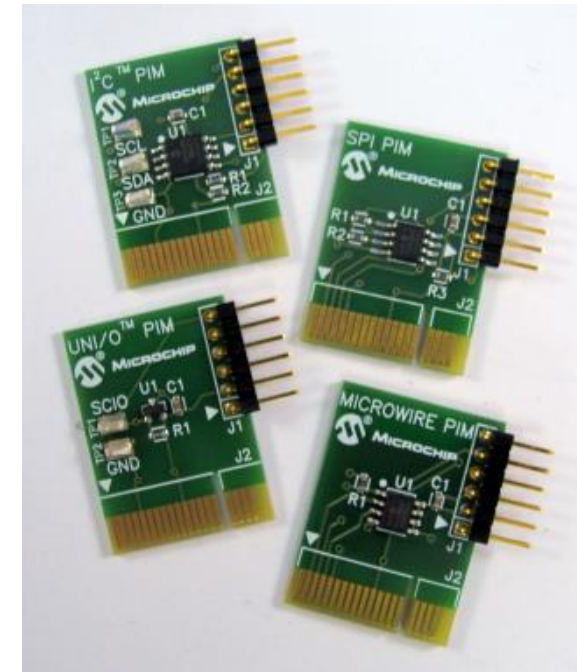
EEPROM PIM PICtail Pack

The Serial EEPROM PIM PICtail™ Modules are a series of boards designed around Microchip Serial EEPROM devices. The boards are designed to interface with the PICtail Plus connector as well as the MPLAB® Starter Kit for Serial Memory Products and the PICkit™ 2.

Features

- Plug-and-play with PICtail™ Plus connector and PICkit™ 2 connector
- High density, wide voltage range devices mounted to the boards
- Follows the published recommended usage for each bus, including decoupling capacitors, pull-up/down resistors
- Test points for oscilloscope connections for firmware debugging (I²C and UNI/O® only)
- All four standard serial interfaces are included for maximum flexibility in developing your application

[Click for More Information](#)

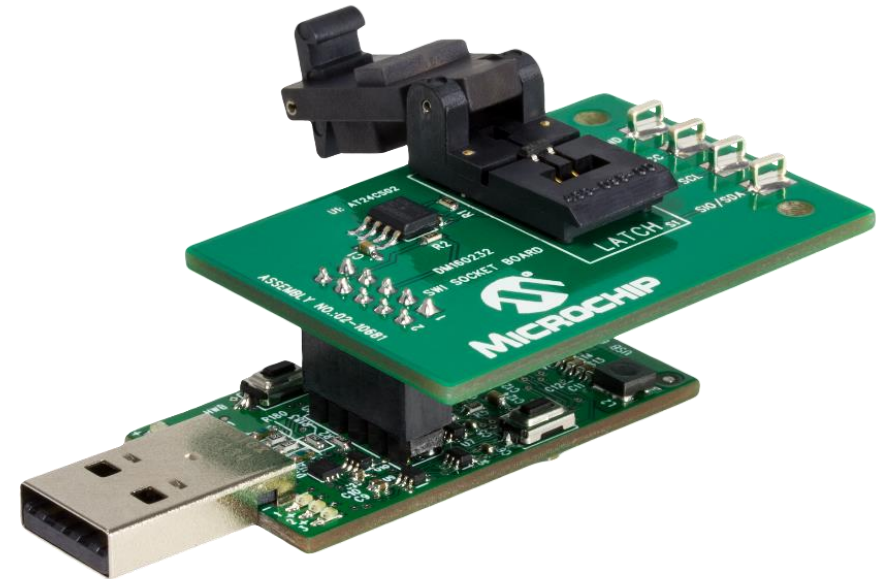


Single-Wire Evaluation Kit

The Serial Memory Single-Wire Evaluation Kit is an easy-to-use interactive user tool to demonstrate the best in class features, functionality and low-power operation of the AT21CS Series Serial EEPROM devices. The evaluation kit is for engineers, developers and decision makers to allow for fast system prototyping using the Single-Wire Interface (SWI) protocol.

Features

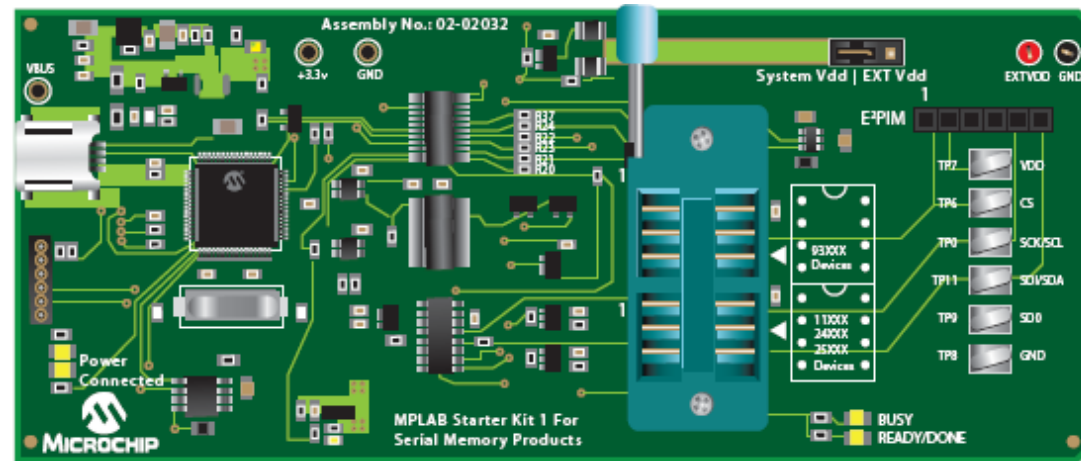
- Includes SWI Graphical User Interface (GUI)
 - Supports Windows®7 or higher operating systems
 - Utilizes USB interface to communicate with user's PC
- Supports USB Base Board firmware update via the Flexible In-system Programming (FLIP) Software Utility
- Variable Socket VPUP Voltage
 - Internal variable voltage from 1.7V to 3.6V



[Click for More Information](#)

MPLAB Starter Kit for Memory Products

Microchip's MPLAB® Starter Kit for Serial Memory Products includes everything necessary to quickly develop a robust and reliable Serial EEPROM design, reducing the time required for system integration and hardware/software fine-tuning.



Features

- 3.3V and 5.0V on-board voltage selection
- 1.8V to 5.5V external voltage support
- Supports Microchip UNI/O® bus, I²C, SPI, and Microwire Serial EEPROMs
- Includes free copy of MPLAB IDE
- USB interconnect

[Click for More Information](#)

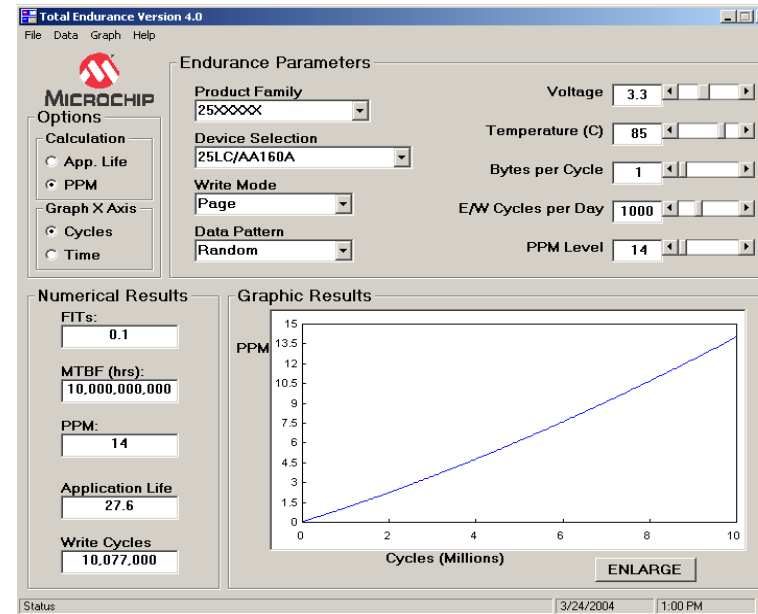
Total Endurance™ Software

Microchip's Total Endurance™ is a simple-to-use, yet powerful, software tool to help system designers determine the endurance of Microchip EEPROMs in a given application. The software will calculate the expected endurance based on the conditions entered by the user, greatly simplifying design trade-off analysis.

Features

- On-screen editing of parameters
- Support of all Microchip Serial EEPROMs

[Click for More Information](#)

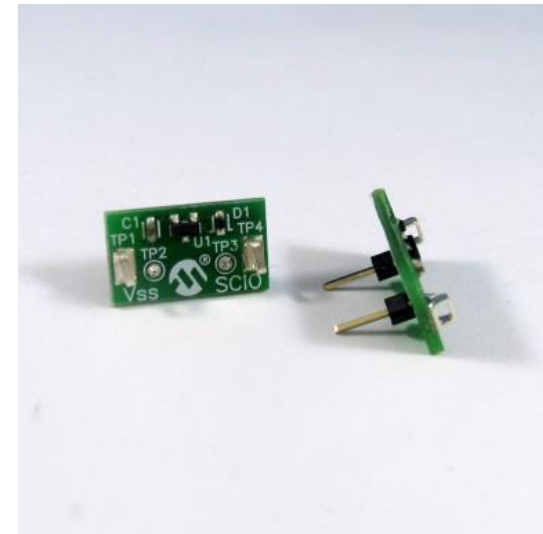


UNI/O[®] Bus Parasitic Power Demo Board

The UNI/O[®] Bus Parasitic Power Demo Board is designed to illustrate how a standard half-wave rectifier and capacitor circuit can be used to parasitically extract power for a UNI/O device from the SCIO signal as described in application note, AN1213 *“Powering a UNI/O[®] Bus Device Through SCIO”*. This reduces the number of connections necessary for adding a UNI/O device to your application down to two: SCIO and Vss.

Features

- 11AA160, 16 Kbit UNI/O Serial EEPROM, featuring 1.8V to 5.5V operation
- Schottky diode and 4.7 μ F capacitor for extracting power parasitically
- Test points for application or oscilloscope connections
- Headers for interfacing with standard .300”-wide DIP sockets
- Can interface with MPLAB[®] Serial Memory Starter Kit



[Click for More Information](#)

SQI SuperFlash® Kit 1

The SQI SuperFlash Kit 1 contains three Serial Flash Daughter Boards. The Serial Flash Daughter Board is an evaluation board designed to interface with the mikroBUS™ connector on the Explorer 16/32 Development Board. It can also interface with the PICtail™ Plus connector that is located on the Explorer 16 Development Board.

Features

- 3-pin header provides easy access for the external power supply and power measurement capability
- Plugs into the mikroBUS connector of the Explorer 16/32 Development Board
- Plugs into the PICtail Plus connector of the Explorer 16 Development Board
- Follows published recommended usage, including decoupling capacitors



[Click for More Information](#)

Serial SuperFlash[®] Kit 2

The Serial Flash Daughter Board is an evaluation board designed to interface with the mikroBUS[™] connector on the Explorer 16/32 Development Board. It can also interface with the PICtail[™] Plus connector located on the Explorer 16 Development Board. The Serial SuperFlash Kit 2 contains three Serial Flash Daughter Boards.

Features

- 3-pin header provides easy access for the external power supply and power measurement capability
- Plugs into the mikroBUS connector of the Explorer 16/32 Development Board
- Plugs into the PICtail Plus connector of the Explorer 16 Development Board
- Follows published recommended usage, including decoupling capacitors



[Click for More
Information](#)

Parallel SuperFlash® Kit 1

The Parallel Flash PICtail™ Plus Daughter Board is designed to interface with the PICtail™ Plus connector found on the Explorer 16 Development Board. The Parallel SuperFlash® Kit 1 contains two Parallel Flash PICtail™ Plus Daughter Boards.

Features

- Contains 2 daughter boards with parallel flash devices SST38VF6401 and SST39VF1601C
- 16-pin header provides easy access to Parallel Flash data pins for probing and evaluation.
- 32-pin right angled header provides easy access to Parallel Flash address and control pins for probing and evaluation.
- 2-pin header/jumper provides flexibility to use either on board power or use external power supply and provides power measurement capability.
- Plugs into PICtail Plus connector of Explorer 16 Development board.
- Follows the published recommended usage including -decoupling capacitors.



[Click for More Information](#)

SPI SRAM PICtail™ with Battery Backup

The SPI SRAM PICtail™ Plus Daughter Board demonstrates the features of the 23LCV1024 1 Mbit Serial SRAM with battery backup. It is designed to interface with the Explorer 16 Development Board or the PICDEM PIC18 Explorer Board.

Features

- Includes 3 V coin cell for battery backup
- PICtail and PICtail Plus connector to interface with all Microchip development boards
- Follows the published recommended usage including -decoupling capacitors



[Click for More Information](#)

I²C EERAM PICtail™ Kit

The EERAM I²C PICtail™ Kit demonstrates the features and abilities of the 47C04 and 47L16 I²C EERAM devices in standard development platforms. By designing this daughter board with the PICtail Plus and mikroBUS™ connectors, it will operate with the Explorer 8 Development Board, the Explorer 16/32 Development Board, and many other tools

Features

- 4 Kbit and 16 Kbit I²C serial EERAM devices
- Follows the published recommended capacitor selection for the EERAM device
- Test points with probe loops for SCL, SDA, VCC and GND
- On-board pull-ups for I²C bus operation
- SMD jumper options for 3.3V and 5V operation
- On-board RC filter ensures VCC ramp rate adheres to data sheet requirements
- 0.1" two-way jumper for VCAP/VCC operation
- PICtail™ Plus and mikroBUS™ connectors



[Click for More Information](#)

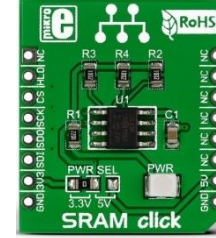
Mikroe Click Boards



2 Mbit, I2C
EEPROM
AT24CM02



64 Mbit, SPI/SQI
NOR Flash
SST26VF064B



1 Mbit, SPI SRAM
23LC1024



2 Kbit with EUI-
64™ MAC I2C
EEPROM
24AA025E64



RTCC, I2C with 1
Kbit EEPROM
MCP79410



RTCC, SPI with 1
Kbit EEPROM
MCP79510

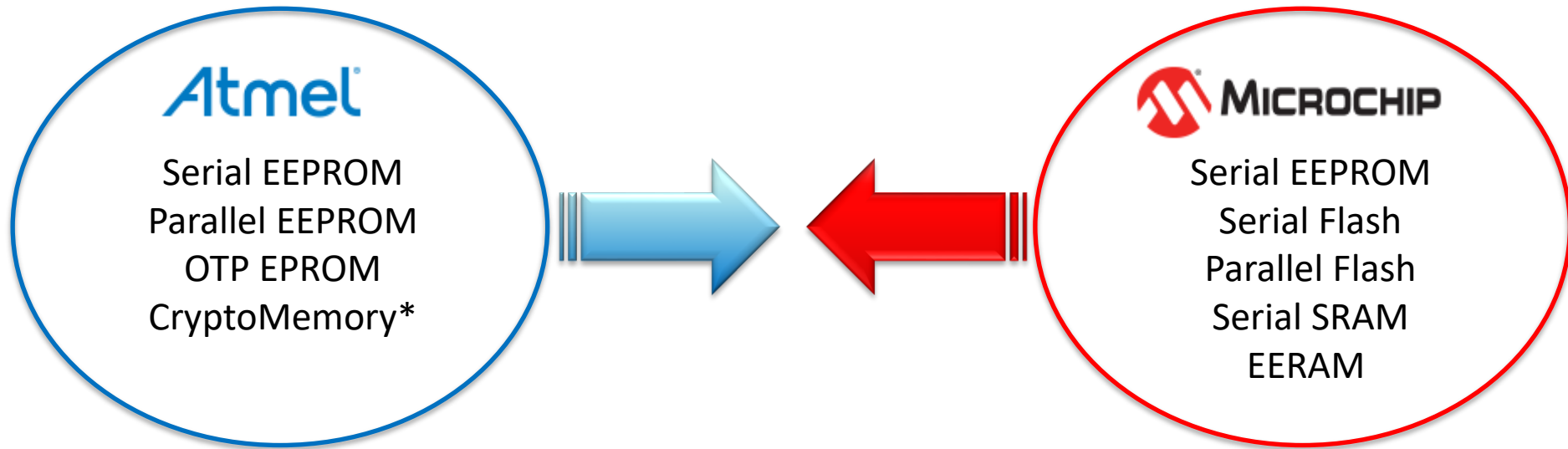


16 Kbit, I2C EERAM
47L16 (3.3V)



16 Kbit, I2C EERAM
47C16 (5V)

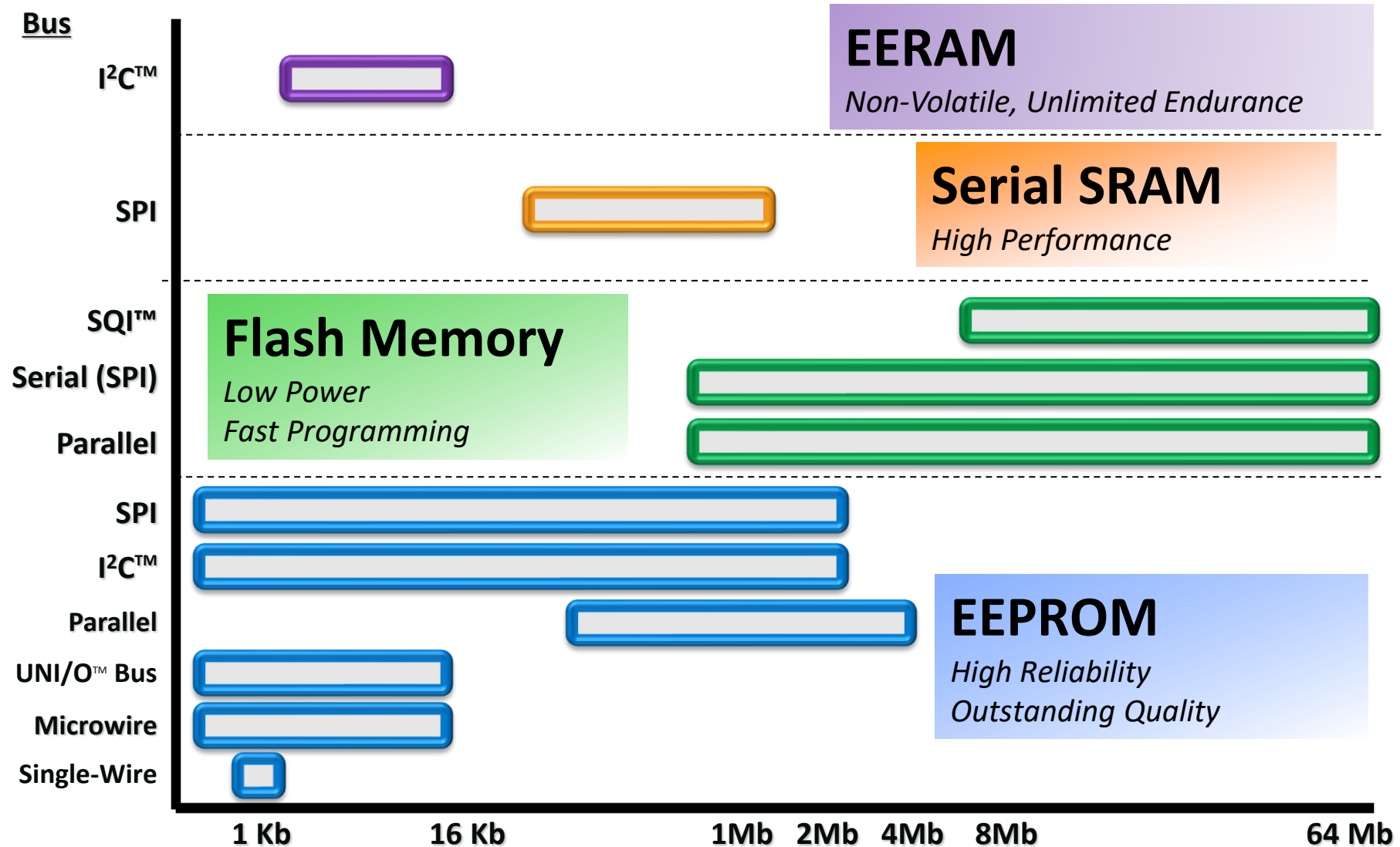
Microchip Memory Overview



- Over 50 years combined experience in EEPROM technology
 - Broadest portfolio of non-volatile memory
 - Market Leadership
- Expanded manufacturing and technology capabilities

**CryptoMemory devices covered by Security Division*

Broad Memory Portfolio



Serial EEPROM Products



Key: In Production In Development Concept



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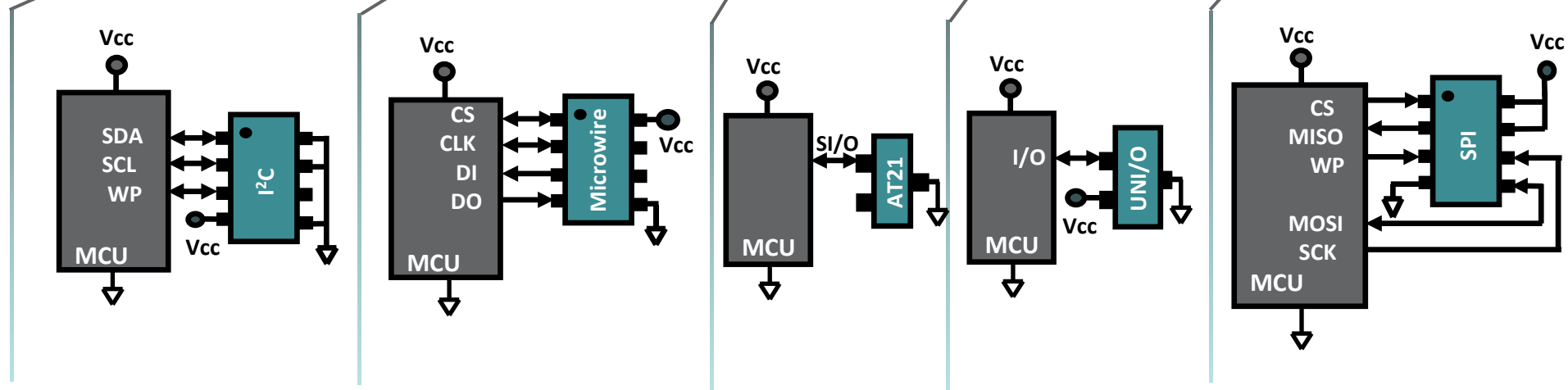
NEXT >>

<< BACK to Products

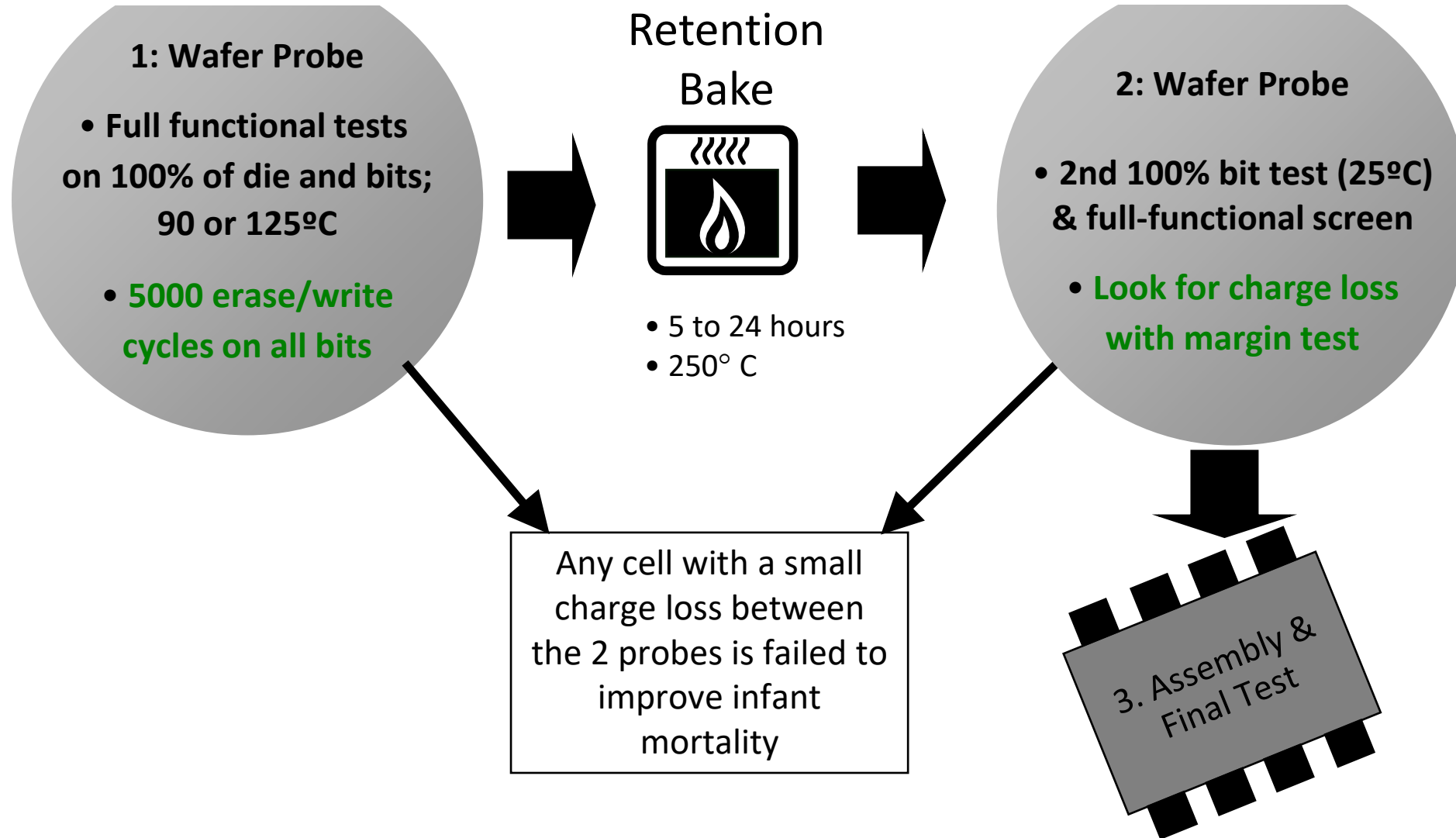


EEPROM Serial Bus Overview

Parameter	I ² C	Microwire	Single-Wire	UNI/O™	SPI
Density Range	128b-2Mbit	1Kb to 16Kb	1Kb	1Kb to 16Kb	1Kb to 2Mbit
Speed	Up to 1 MHz	Up to 3 MHz	up to 125kbps	Up to 100 KHz	Up to 20 MHz
I/O Pins	2: Clock, Data	4: Clock, CS, DI, DO	1: SI/O	1: Clock/Data	4: SCK, CS, DI, DO
Write Protect	HW, SW	HW	SW	SW	HW, SW
Pricing	Least Expensive				Most Expensive

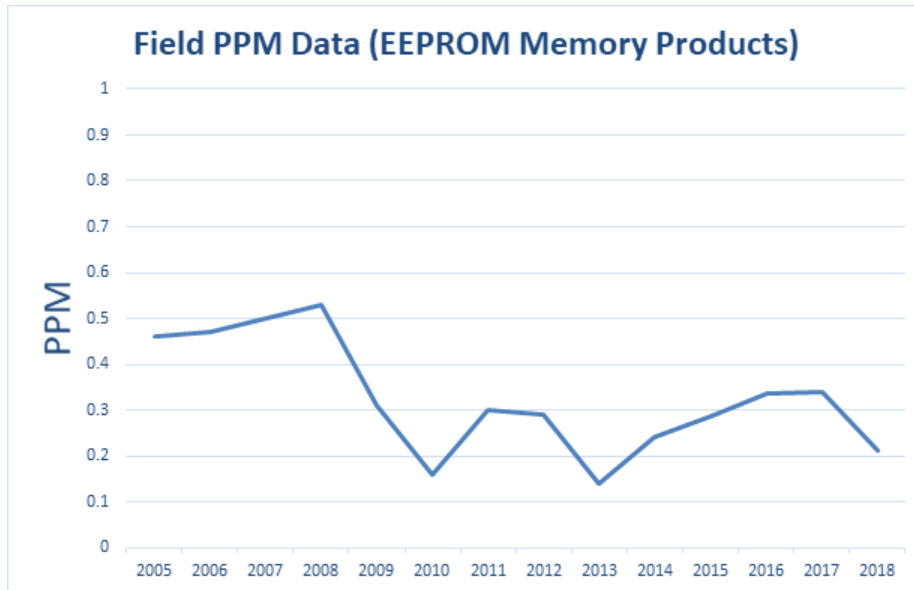


Triple-test Strategy: Reducing Fails



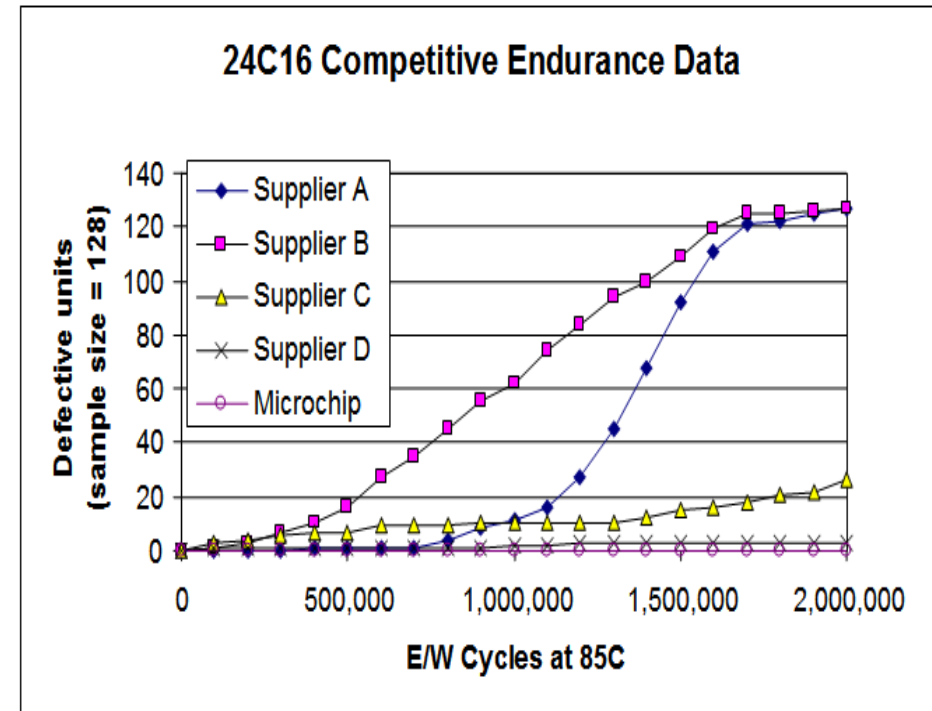
2 Goals:

Zero Infant Mortality



Great Testing

Best Long-Term Endurance



Great EEPROM cell

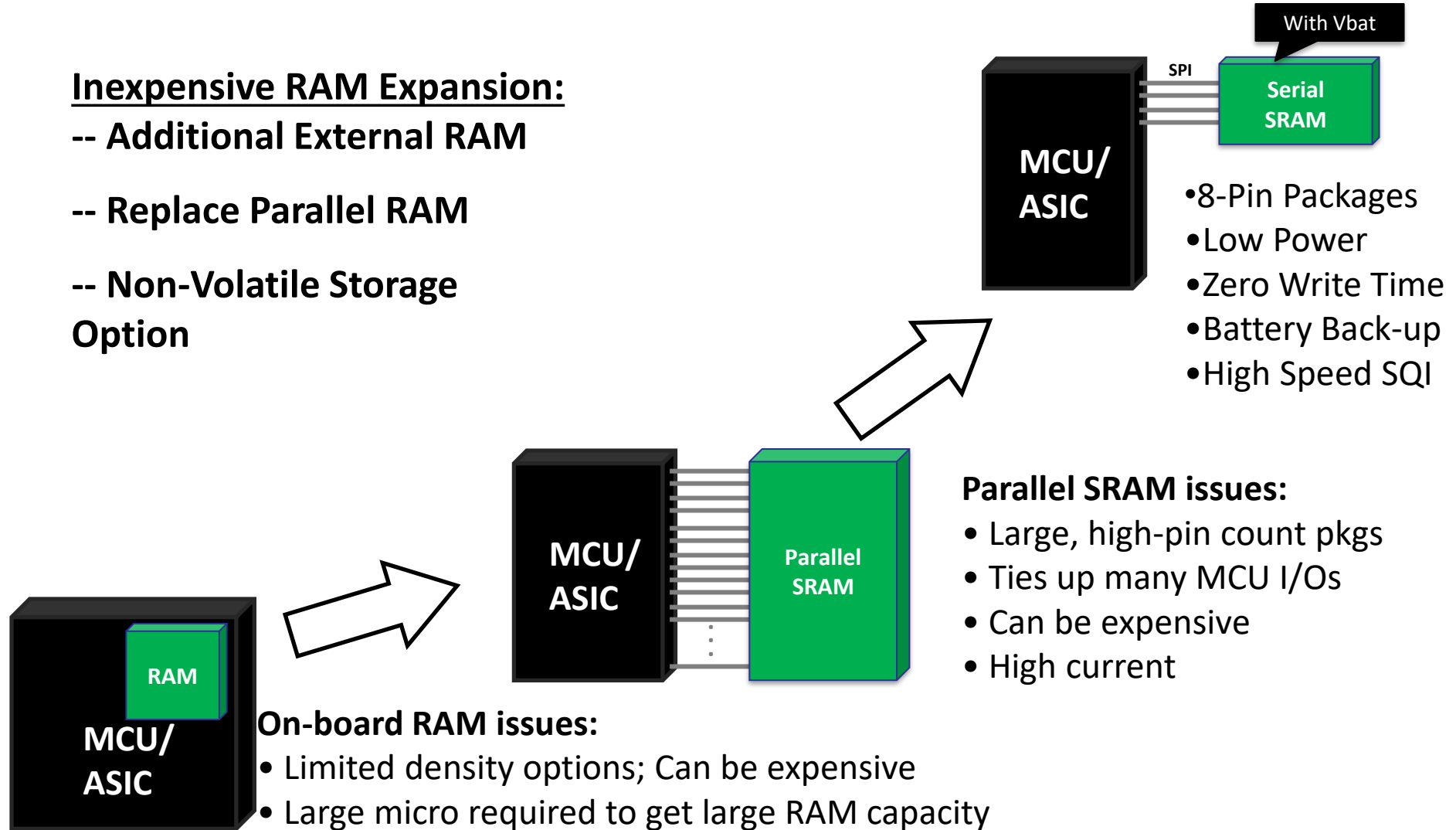
EEPROM Applications

	Microchip Strengths	Applications
Automotive	Quality, PPAP, long product life cycles, AEC Q-100 & TSO-16949 compliant, in-house testing, triple-test flow	Transmission, ABS, Power Train, Airbag, Entertainment System, Mirror Controllers, GPS, Steering Control
Medical	Outstanding quality, and reliability, tiny packages, low power, Microchip owned fabs	Blood Glucose Meters, Hearing Aid, Medical Imaging Devices, XRAY, Oxygen Concentrator, ECG
Industrial	Robust designs, high endurance, wide operating range, custom part options	Metering, Industrial Control, Security Systems, Alarm Systems, Sensor Networks, Battery Chargers
Consumer	Short lead times, excellent supply, innovative packages, application specific options, low power	LCD TV, Set Top Box, Printers, Blue Tooth, Wireless, Laptop, Mouse, Networking, DDR2/3, Camera, POS, Appliances

Serial SRAM: Inexpensive RAM Expansion

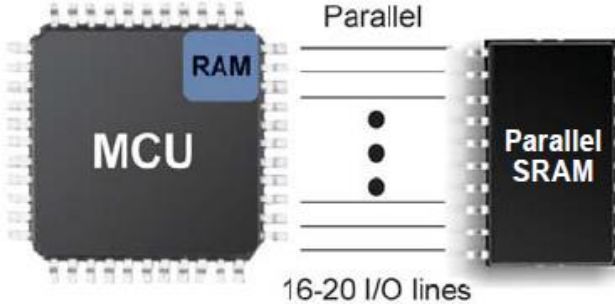
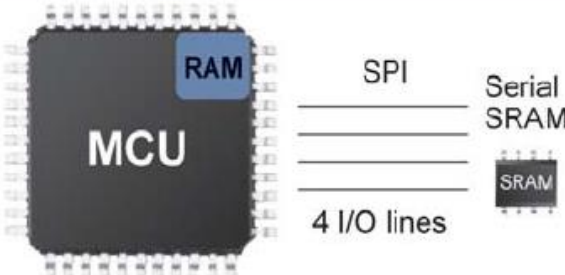
Inexpensive RAM Expansion:

- Additional External RAM
- Replace Parallel RAM
- Non-Volatile Storage Option



Serial SRAM Overview

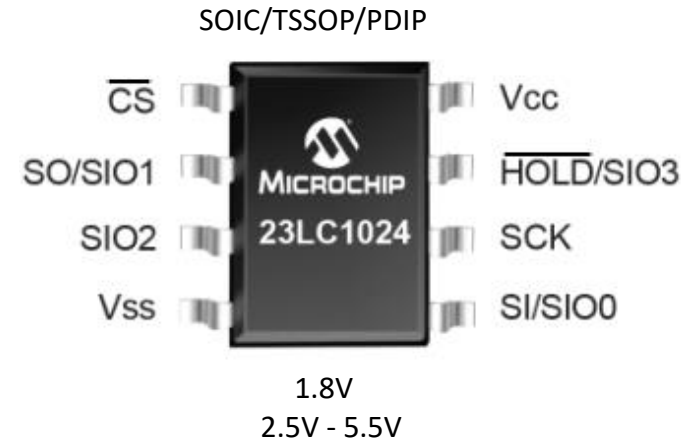
The Serial Advantage – Not Just Price

Feature	Traditional Parallel SRAM	Microchip Serial SRAM
I/O Connection to MCU	16-20	4
Standby Current	3 mA	1 μ A
Active Current	50 mA	1-10 mA
Lowest Operating Voltage	3.0V	1.7V
Footprint	100 mm ²	20 mm ²
Smallest Packages	28-TSSOP, 28-SOIC	8-TSSOP, 8-SOIC
 MCU RAM  Parallel MCU RAM Parallel SRAM 16-20 I/O lines  SPI MCU RAM Serial SRAM SRAM 4 I/O lines Stand-alone serial SRAM offering greater design flexibility and the opportunity for RAM expansion		

Serial SRAM Overview

Serial SRAM

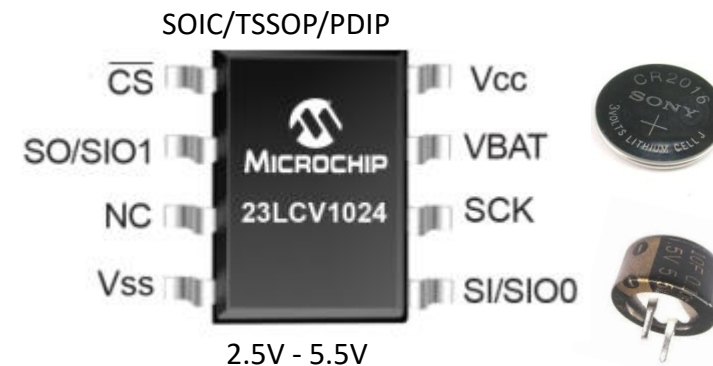
- 64Kb – 1Mb
- SPI/SDI*/SQI* Modes – Up to 4x speeds
- 20MHz Clock Rate (SPI)
- Unlimited Endurance
- Zero Write Times
- Supports 5V*



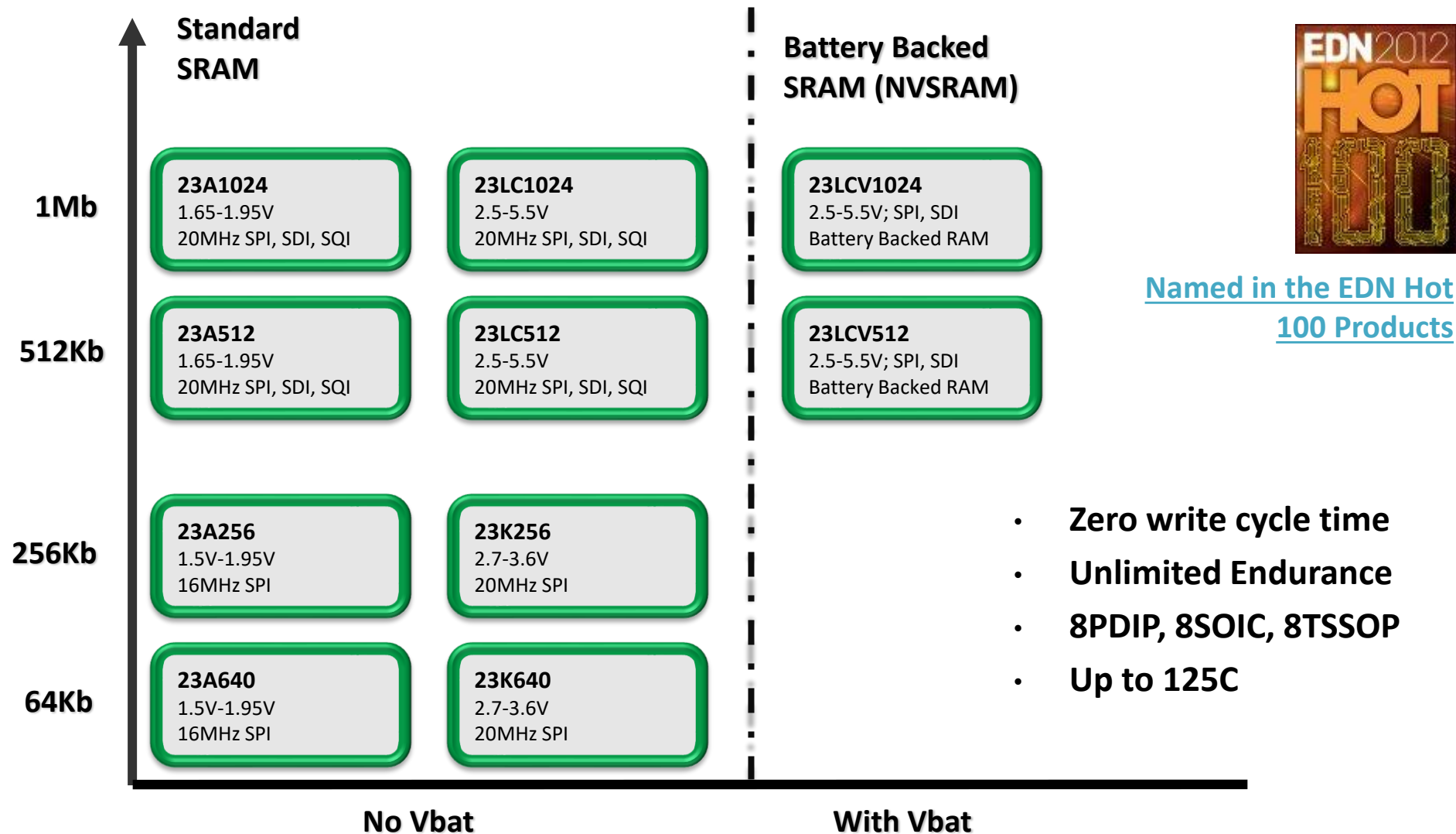
*Only on 512Kb and 1Mb devices

Serial NVSRAM – Battery Backed SRAM

- 512Kb – 1Mb
- Automatic Battery Backup via V_{BAT}
- 20+ Year Data Retention (CR2032)
- 1.5V Vtrip
- Unlimited Endurance
- Supports SPI and SDI modes (no SQI)



Serial SRAM Family



Serial SRAM Applications

- **Buffering:**
 - Printers
 - Wireless Modules
 - ATM
 - POS
 - Internet (VoIP, Internet Radio)
- **Endurance: (NVSRAM)**
 - Metering
 - RAID Systems
 - Vehicle Black Box
 - Data Recorders
- **Data Intensive Functions:**
 - Voice and Audio
 - Graphics (LCD Drivers)
 - Camera Modules in Cars
 - Math Intensive Functions

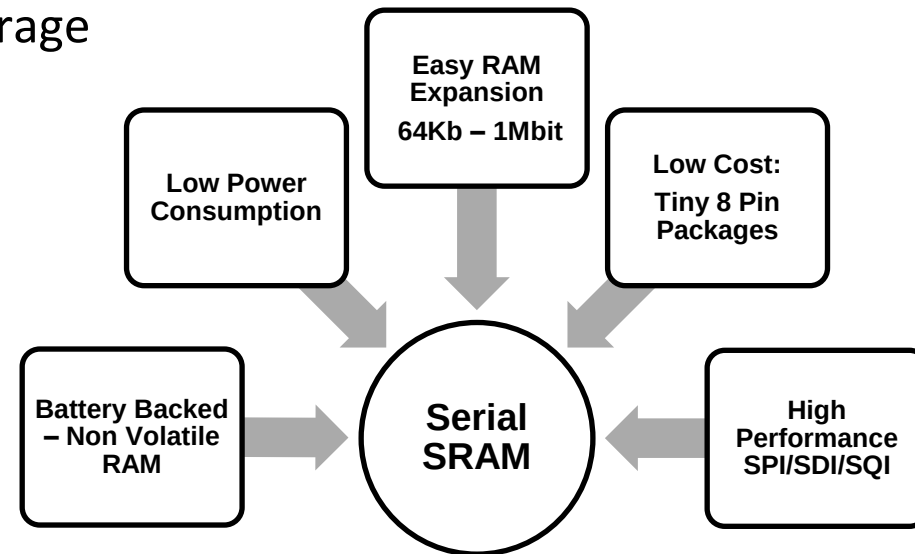
Applications:

- Set-top boxes
- Printers
- ATM, Kiosks
- Metering
- POS terminals
- Slot Machines
- Industrial automation
- RAID controllers
- Data communications
- Multi-function printers
- Single board computers
- Data Recorder/Loggers

Serial SRAM Summary

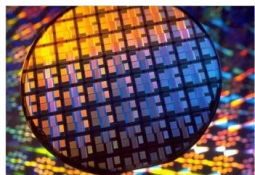
Offers the *least expensive solution* for customers and designers looking for more RAM in their application.

- Easy RAM Expansion - Up to 128Kbytes
- Lowest Cost RAM Solution available
- Low Power Draw vs. Parallel SRAM
- High Performance with SDI and SQI (4x Speed)
- Battery Backup Capability – Nonvolatile Storage
- Unlimited Instantaneous Writes
- Supports 5V



NOR Flash Overview

- ❖ **SuperFlash® Technology**
 - ❖ High Reliability - 100 Years Data Retention
- ❖ **Fast program and erase times**
- ❖ **Small sector architecture**
- ❖ **Low power**
- ❖ **Broad product offering**
 - Parallel and Serial Flash
 - 1.8 – 5.0V
 - 512Kb – 64Mb
- ❖ **Industry's smallest packages**



Die / Wafer



8-WLCSP



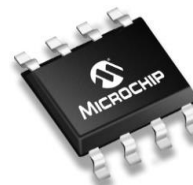
8-USON



8-MSOP



8-WSON



8-SOIC



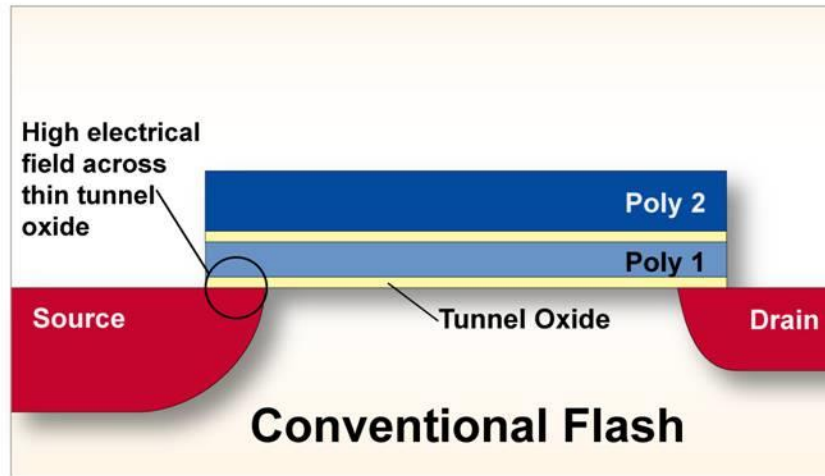
16-SOIC



24-TFBGA

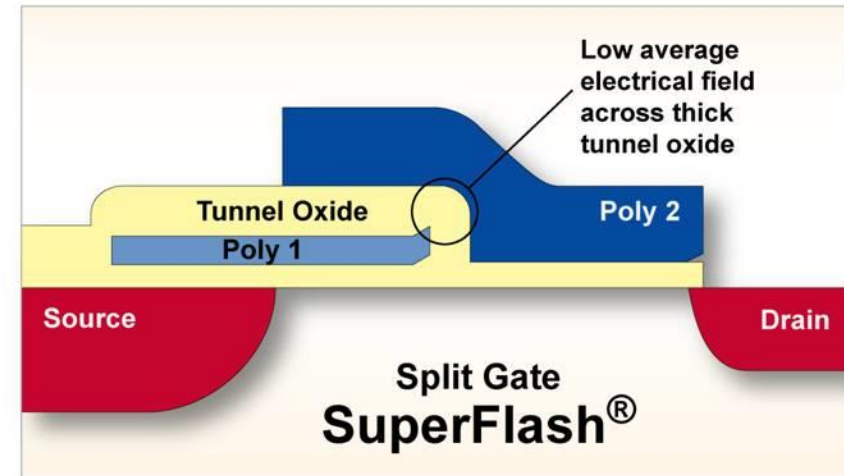
Advantages of SuperFlash[®] Technology

Conventional Flash



Symmetrical trap generation → Charge Loss

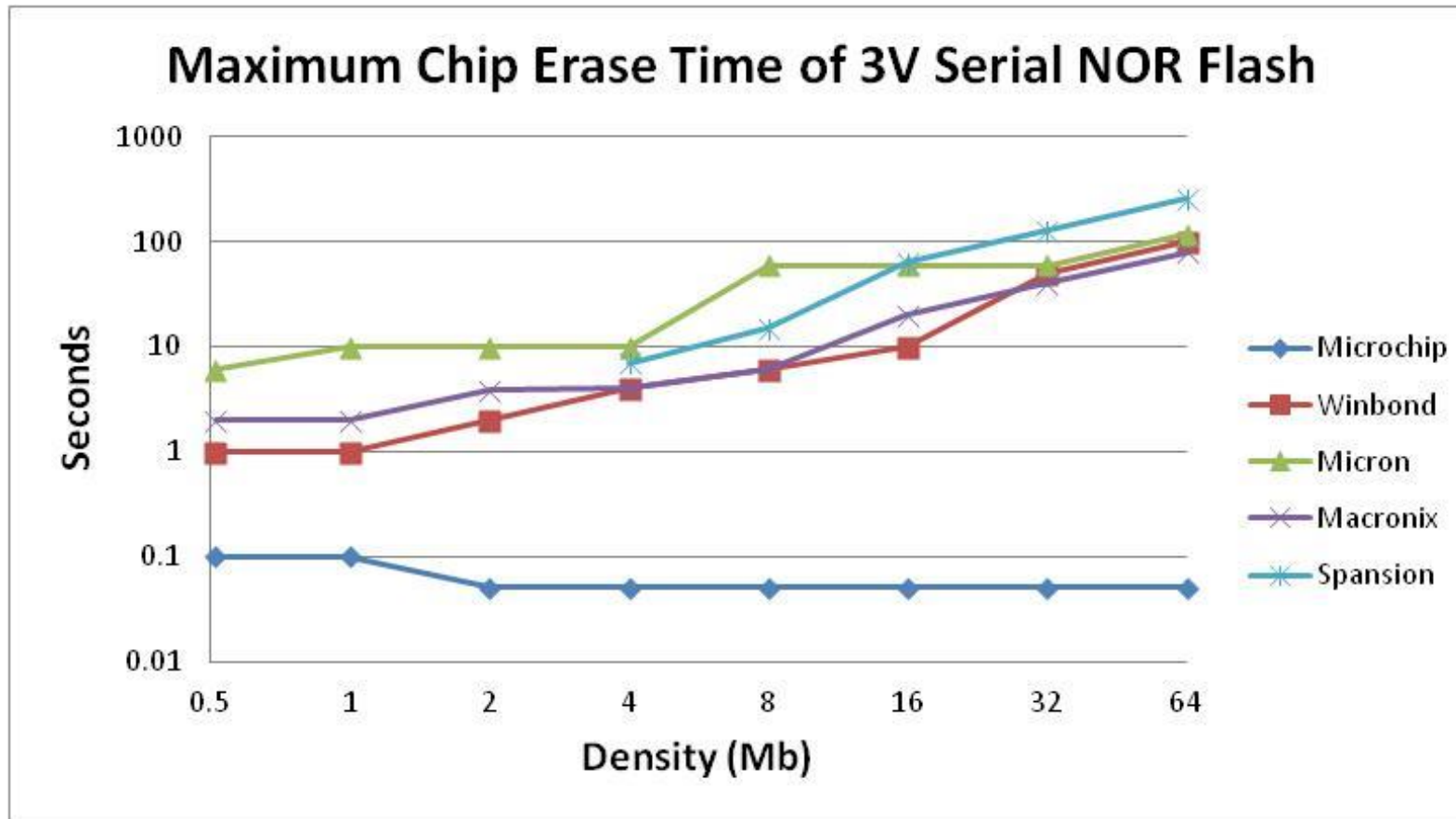
SuperFlash[®]



Asymmetrical trap generation → No Charge Loss

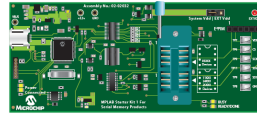
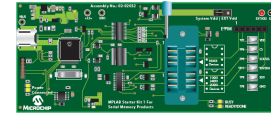
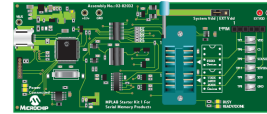
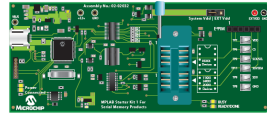
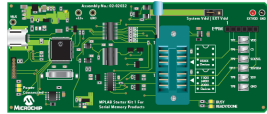
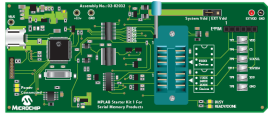
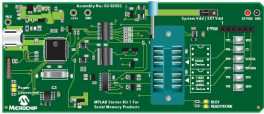
- Superior Reliability
- Low-voltage, Low-power Operation
- Fast, Fixed Program & Erase Times

1,000x Faster Erase Performance

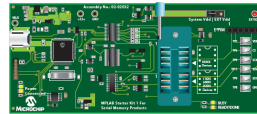
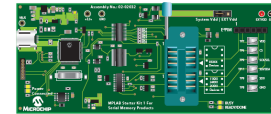
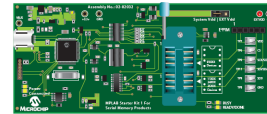
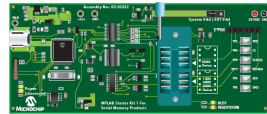
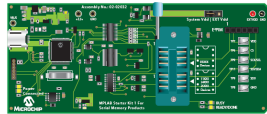
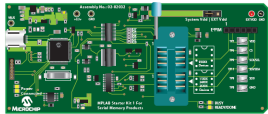
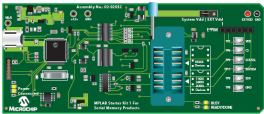


**Competition's long chip erase time increases cost
and creates bottleneck during manufacturing**

Faster Production Throughput



Slow Production Testing with Stacked Gate Flash



Super Fast Production Testing with SuperFlash

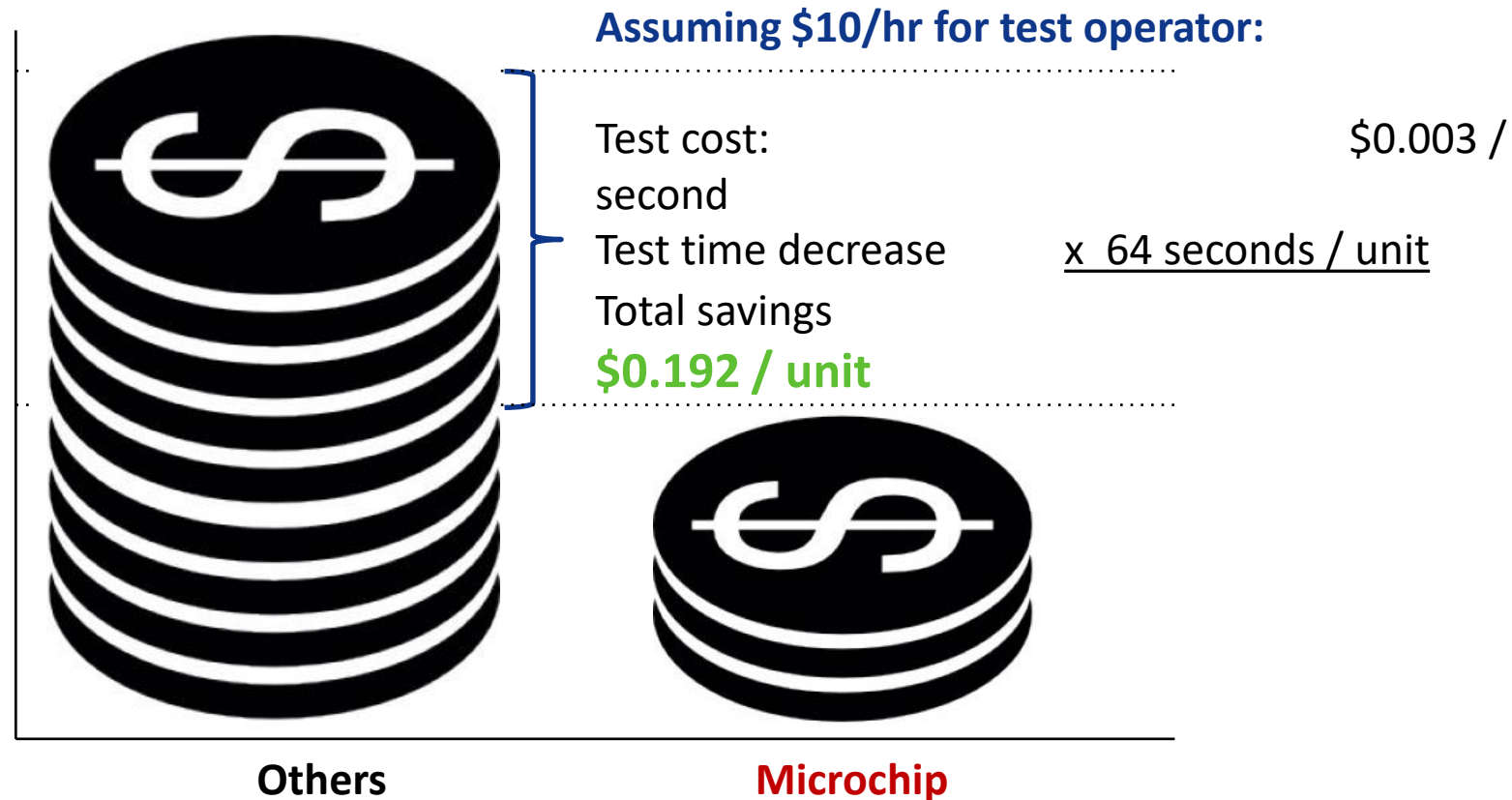


Saves time and Money in Your Production Line!

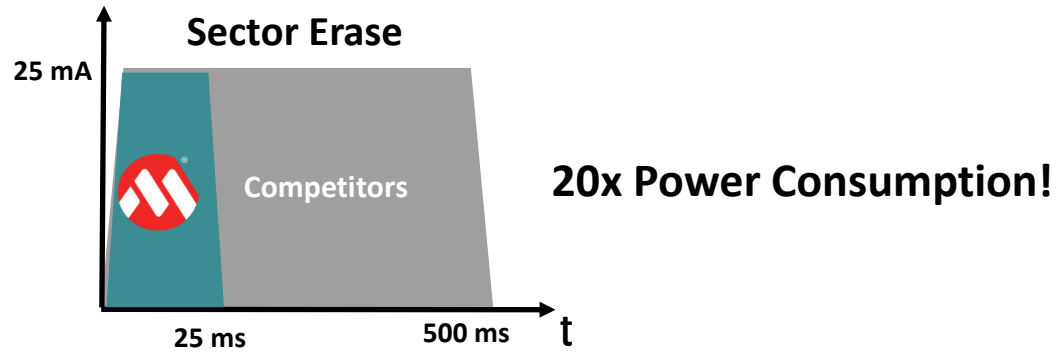
Time is Money

Short test time brings lower total cost of ownership

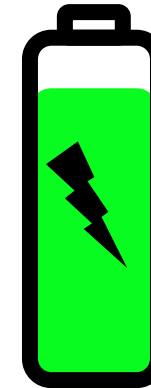
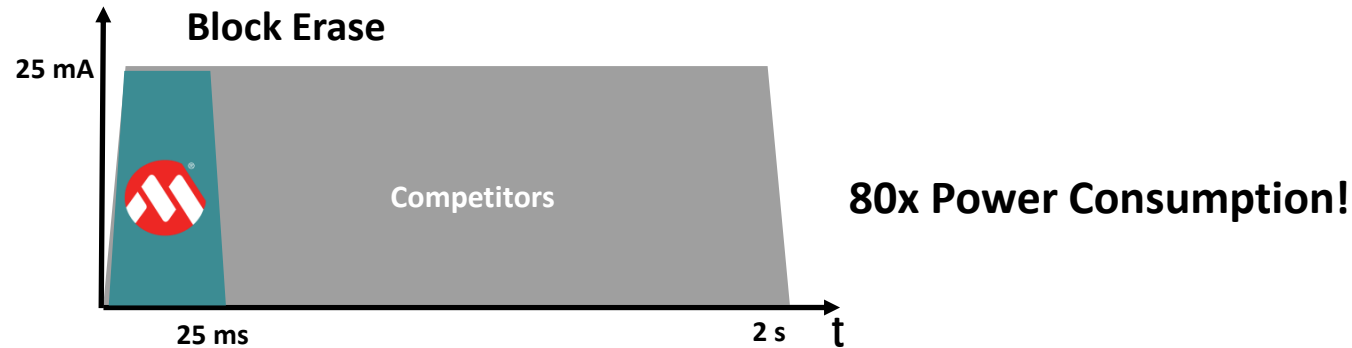
- ❑ Industrial and automotive products require extensive testing in production
- ❑ Real-life case study



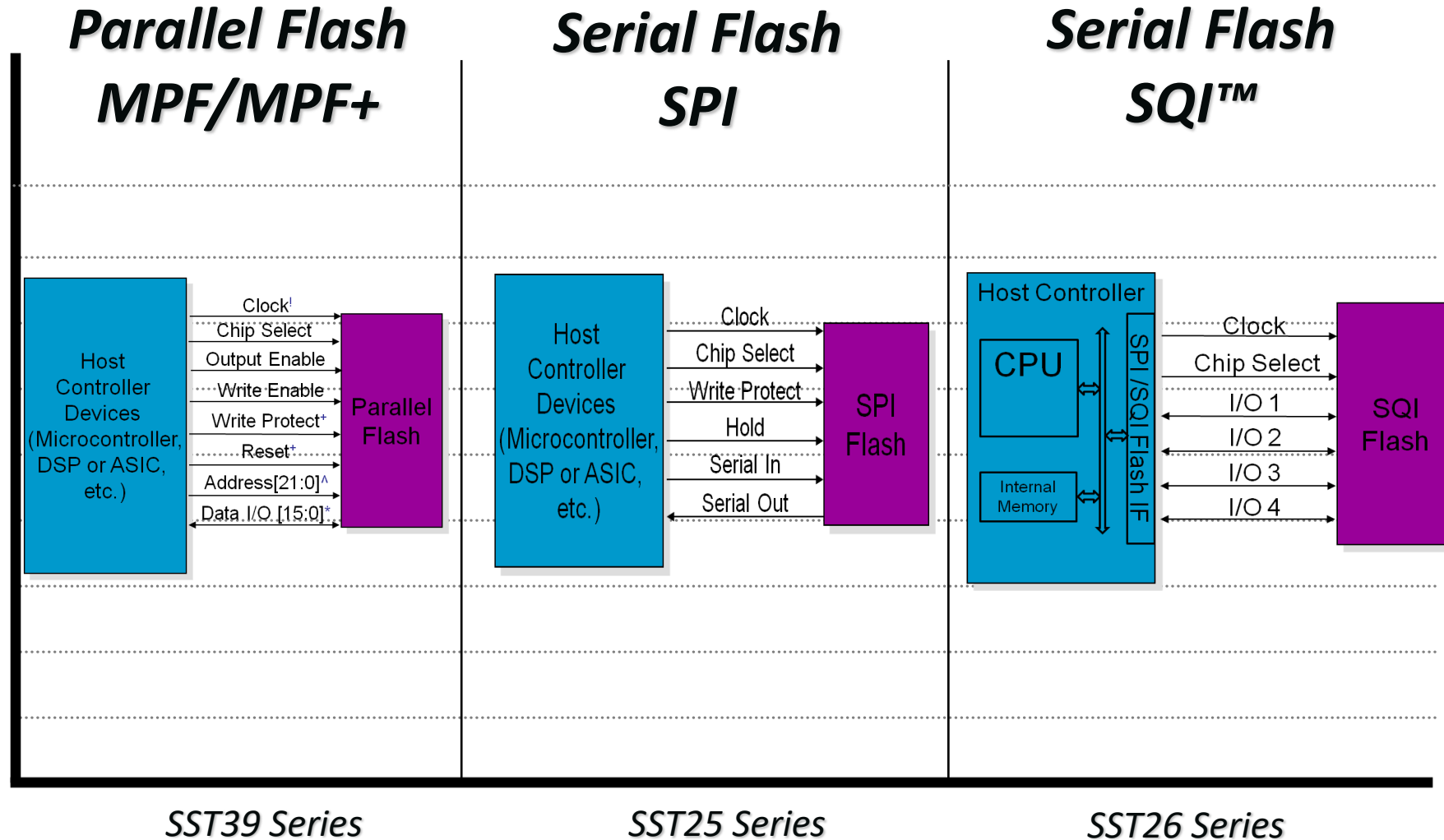
Power Consumption During Erase



Datasheets specs for I_{ccw} are typically 25mA, however.....



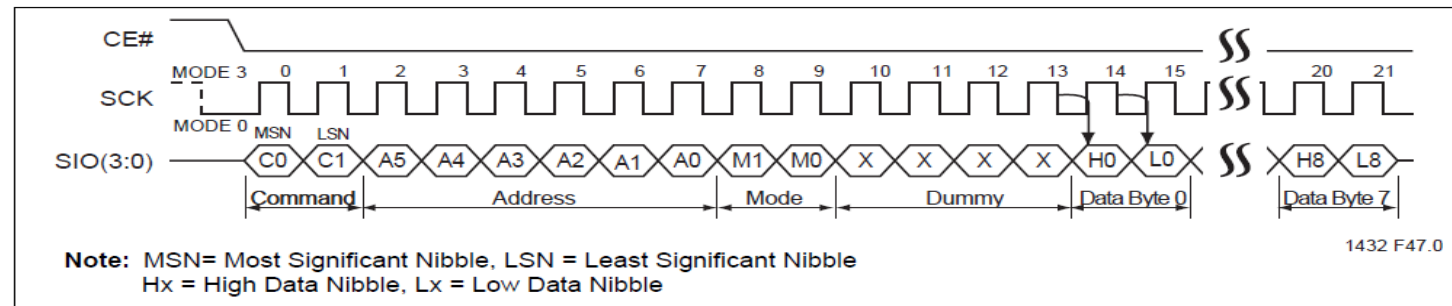
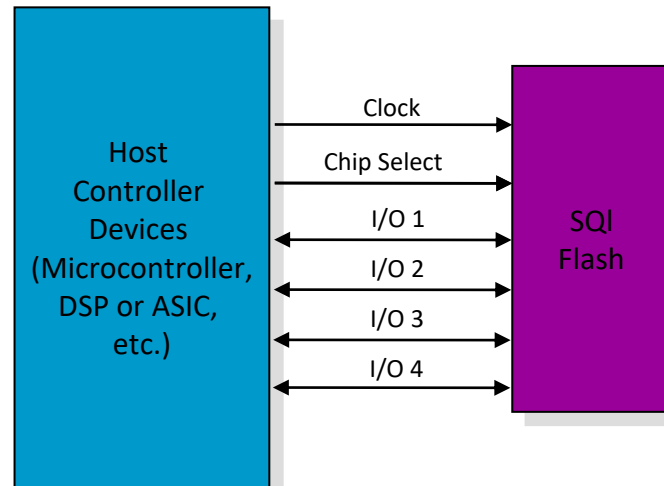
NOR FLASH Interfaces



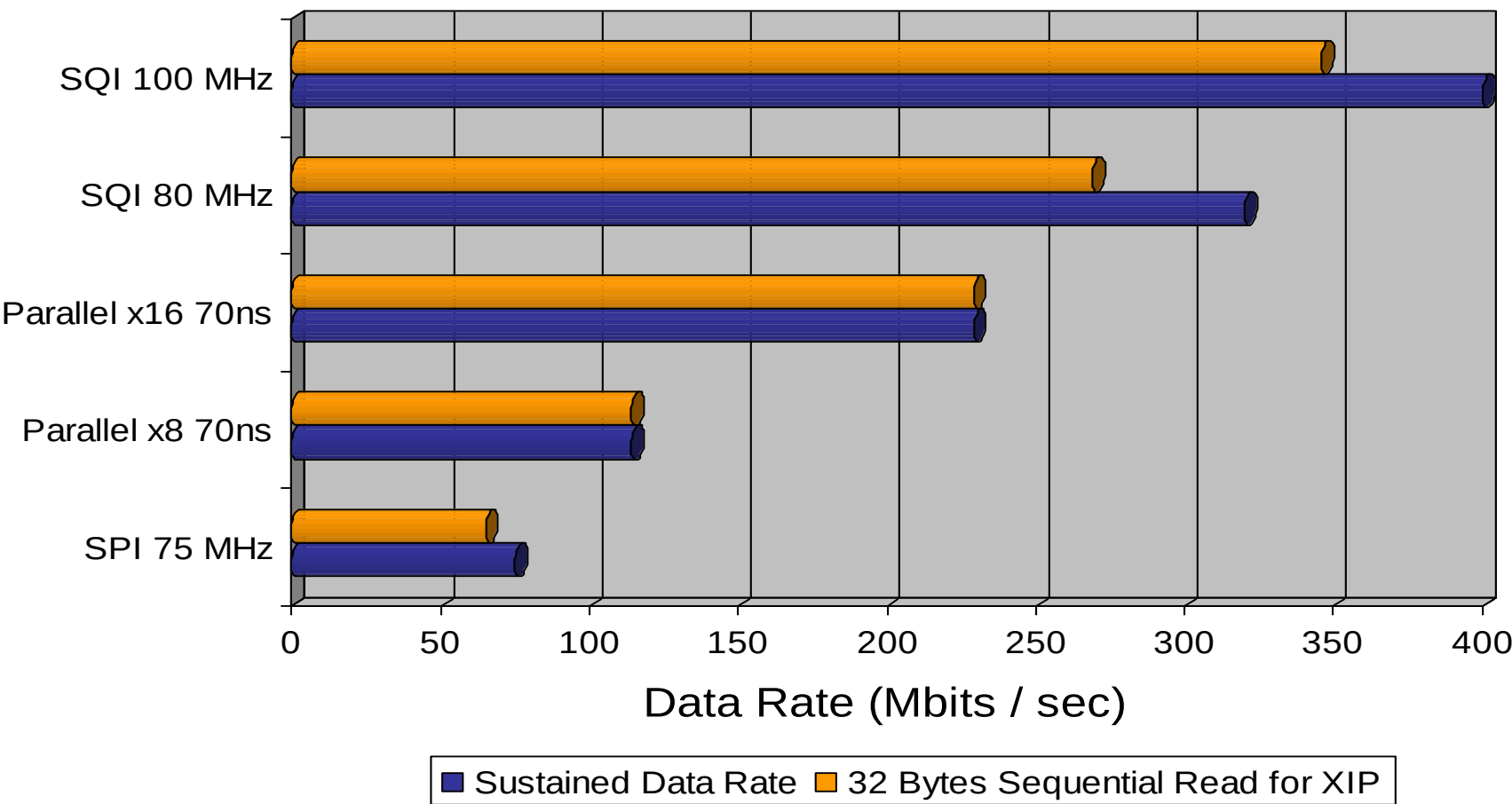
What is SQI?

- **SQI flash is Microchip Trademark**

- ❖ A flash memory that reads and writes through a 4-bit multiplexed synchronous serial communication interface
- ❖ Enables true low-pin count, high-bandwidth code XIP applications



The Bandwidth Dilemma



26 Series SQI Flash Devices Feature Fast Operating Frequency and I/O Architecture Enhancements for Execute-in-Place Capability

NOR FLASH Applications



Connected Home

- Set-top box media central “hub”
- Home server/multimedia storage
- Wireless media distribution
- Content uploads to PND/Automobile
- Internet accessible via multiple networks
- Network Appliances



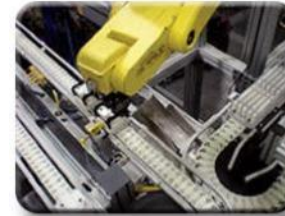
Enterprise

- Data Center
- Banking Mainframe Servers
- Cloud-based Servers
- Enterprise Storage Solutions
- High Speed Optical Networks



Medical

- Wireless bio-monitoring/diagnostics
- Implantable devices
- Self-dispensing medications
- Personal diagnostic equipment
- Patient-to-medical database bridge



Industrial

- Portable inventory control
- Wireless networking
- Advanced security monitoring
- Machine-to-machine communication
- Smart Meters

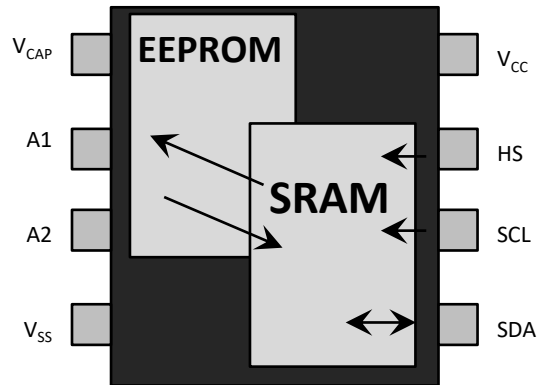


Automotive

- Home-to-vehicle multimedia link
- Infotainment
- Multimedia wireless docking station
- Location-based server

What is EERAM?

EERAM is an SRAM with a shadow EEPROM in one package

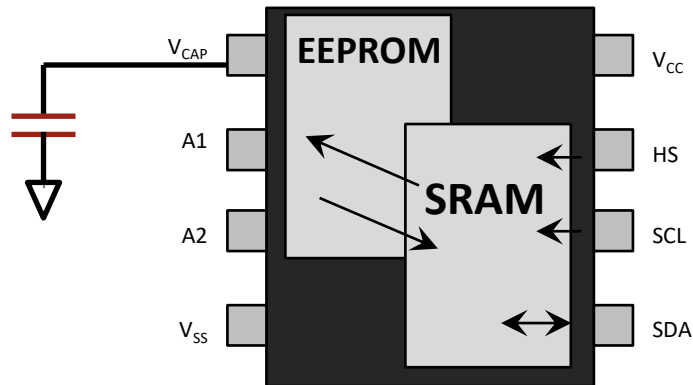


- 3.0V (SPI, I²C); 5.0V (I²C)
- Unlimited Writes to SRAM
- Automatically Stores Data on power down
- No Battery Needed (needs ext. capacitor)
- Data auto recalled to SRAM on Power-Up

Combines Two Proven Technologies
***“Reliability of an EEPROM with the
Performance of an SRAM”***

What Problem does it Solve?

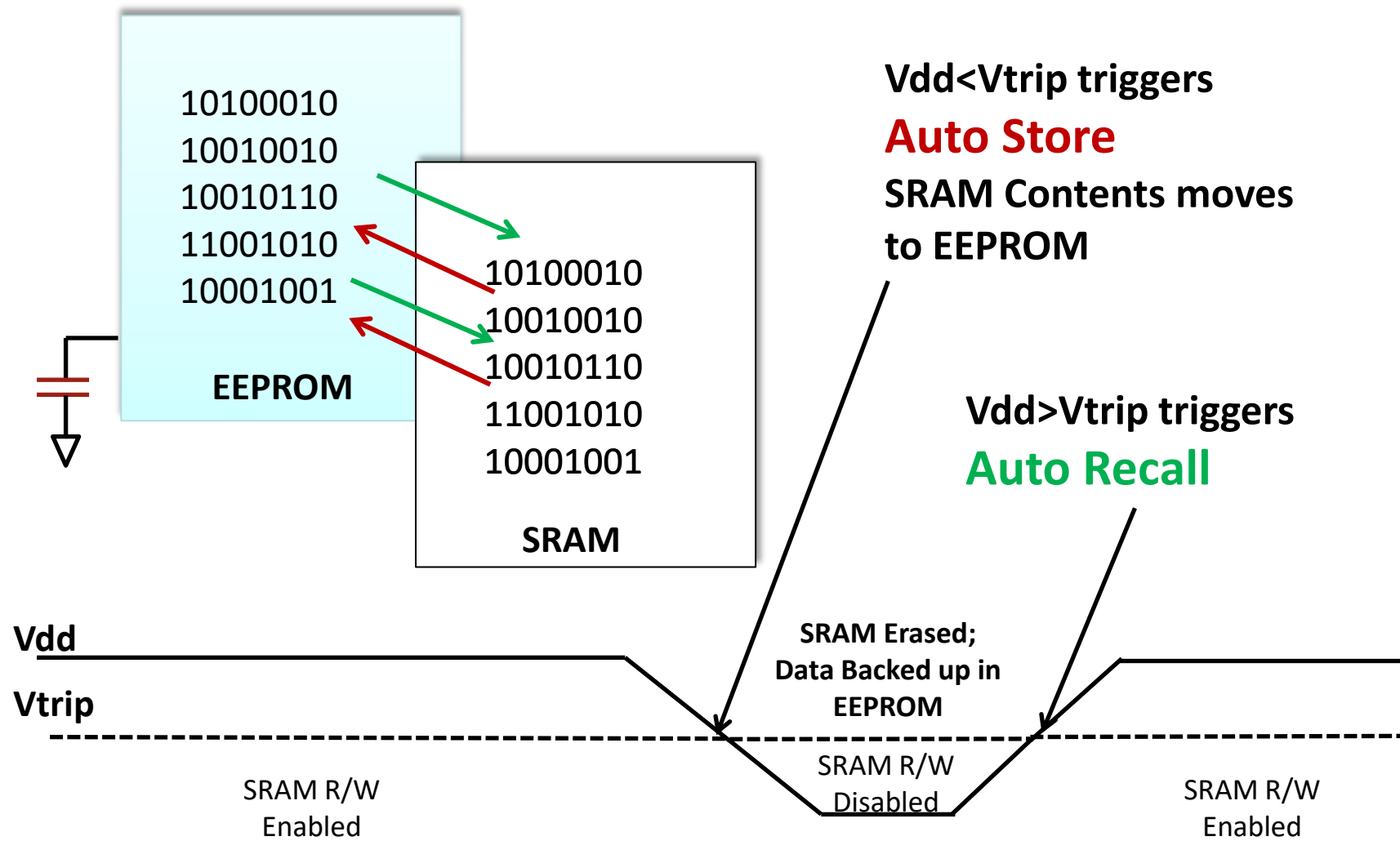
- For applications that need data to be constantly updated, **EERAM offers a safe way to automatically and safely store data during a power loss event.**



The capacitor connected to the Vcap pin provides the necessary energy to safely copy RAM contents to secure EEPROM back up during power loss.

Upon Power Up data is recalled automatically from EEPROM to SRAM

How does EERAM Work?



Benefits of EERAM



Preserve Data Reliably Through Power Loss



No Battery Needed



Unlimited Writes to Array



Lowest Cost Solution



Zero Write Times, Random Access

Applications



Metering – Energy, Gas, Water

- Secure, Continuous logging of consumption data
- In field updates of features and pay scale tables



Automotive

- Black Box Data Recorder, Data Logging
- Last Set Point Recorder (Seat Position, Wiper etc)
- ABS, Air Bags, Sensors, Seats, Black Box, Transmission



Printers, ATM, Kiosks, POS

- Log Printer Head Position, Record Ink Details
- Buffer Memory
- Record Number of Prints, Start Up Configuration

Any application that needs **unlimited writes**, **instant writes**, and **data stored securely during power loss** is a good fit for EERAM